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SYSTEM FOR SHAPING AND IMPLANTING BIOLOGIC INTRAOCULAR STENT FOR INCREASED AQUEOUS OUTFLOW AND LOWERING OF INTRAOCULAR PRESSURE

Abstract

Systems and methods for preparation of an implant and for ab interno insertion of the implant into an eye of a patient including a tissue cartridge configured to receive and hold a patch of a material; a cutting device; and a delivery device. Related devices, systems, and methods are provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 17/325,785 filed May 20, 2021, which claims the benefit of priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. Nos. 63/027,689, filed May 20, 2020, and 63/163,623, filed Mar. 19, 2021. The disclosures of the applications are incorporated by reference in their entireties.

BACKGROUND

[0002] The mainstay of ophthalmic surgery for glaucoma is the enhancement of aqueous outflow from the eye. There are various approaches to such surgery, including: 1) ab externo trabeculectomy or shunting, which requires cutting the conjunctiva and the sclera to penetrate the eye and provide a trans-scleral outflow path; 2) ab interno trabecular or trans-scleral outflow stenting or shunting of aqueous with hardware-based implantable devices or with ablating, non-implantable cutters such as dual-blade and trabectome; and 3) ab interno supraciliary stenting using implantable non-biological hardware implants.

[0003] Current ab interno stenting devices and methods are based on non-biological hardware materials such as polyimide, polyethersulphone, titanium, poly styrene-blocks-isobutylene-block-styrene and others. There are significant drawbacks with such non-biological hardware-based implantable devices as such devices can lead to major erosion, fibrosis and ocular tissue damage such as endothelial cell loss.

[0004] In view of the foregoing, there is a need for improved devices and methods related to ophthalmic surgery for the treatment of glaucoma.

SUMMARY

[0005] In an aspect, described is a system for preparation of an implant and ab interno insertion of the implant into an eye of a patient. The system includes a tissue cartridge configured to receive and hold a patch of a material; a cutting device; and a delivery device.

[0006] The tissue cartridge can include a shaft extending from a distal end of the tissue cartridge. At least a distal end region of the shaft is sized and shaped for insertion into an anterior chamber of the eye. The shaft can include a lumen. The tissue cartridge can further include a base and a cover. The base can be configured to receive the patch and the cover configured to hold the patch fixed against the base. The cutting device can include a cutting member configured to cut the patch of a material positioned within the tissue cartridge. Cutting the patch of a material with the cutting member can form an implant from the patch. The implant can be configured for implantation into the eye of the patient. The delivery device can include an actuator configured to deploy the implant positioned within the cartridge through the lumen of the shaft into the eye.

[0007] In an interrelated implementation, described is a method of preparing an implant for implantation into, and of inserting said implant into, an eye of a patient. The method includes inserting a patch of a material into a tissue cartridge. The tissue cartridge includes a shaft extending from a distal end of the tissue cartridge. At least a distal end region of the shaft is sized and shaped for insertion into an anterior chamber of the eye. The shaft includes a lumen. The method further includes coupling the tissue cartridge with a cutting device. The cutting device has a cutting member configured to cut the patch of a material within the tissue cartridge. The method further includes cutting the patch with the cutting member to form the implant from the patch while the tissue cartridge is coupled with the cutting device; decoupling the tissue cartridge from the cutting device; coupling the tissue cartridge to a delivery device; inserting the distal end region of the shaft into the anterior chamber of the eye; positioning the distal end region adjacent eye tissue; and actuating the delivery device to deploy the implant from the cartridge through at least a portion of the lumen such that the implant engages the eye tissue. The method can further include delivering a

viscous material through the shaft.

[0008] In an interrelated implementation, described is a system for preparation of an implant and ab interno insertion of the implant into an eye of a patient. The system includes a tissue cartridge configured to receive and hold a patch of a material; and a delivery device.

[0009] The tissue cartridge can include a shaft extending from a distal end of the tissue cartridge. At least a distal end region of the shaft can be sized and shaped for insertion into an anterior chamber of the eye. The shaft can include a lumen. The tissue cartridge can further include a base and a cover. The base can be configured to receive the patch and the cover configured to hold the patch fixed against the base. The system can further include a cutting device. The cutting device can include a cutting member configured to cut the patch of a material positioned within the tissue cartridge. Cutting the patch of a material with the cutting member can form an implant from the patch. The implant can be configured for implantation into the eye of the patient. The delivery device can include an actuator configured to deploy the implant positioned within at least a portion of the cartridge through the lumen of the shaft into the eye. The tissue cartridge can include a nose cone assembly having the distal end region of the tissue cartridge and the shaft. The nose cone assembly can be reversibly coupled to the tissue cartridge and reversibly coupled to the delivery device. The shaft of the tissue cartridge can be configured to deliver a viscous material.

[0010] In an interrelated implementation, described is a method preparing an implant for implantation into, and of inserting said implant into, an eye of a patient. The method includes inserting a patch of a material into a tissue cartridge. The tissue cartridge includes a shaft extending from a distal end of the tissue cartridge. At least a distal end region of the shaft is sized and shaped for insertion into an anterior chamber of the eye. The shaft includes a lumen. The method includes coupling the tissue cartridge with a cutting device. The cutting device has a cutting member configured to cut the patch of a material within the tissue cartridge. The method includes cutting the patch with the cutting member to form the implant from the patch while the tissue cartridge is coupled with the cutting device; decoupling at least a portion of the tissue cartridge from the cutting device; coupling the at least a portion of the tissue cartridge to a delivery device; inserting the distal end region of the shaft into the anterior chamber of the eye; positioning the distal end region adjacent eye tissue; and actuating the delivery device to deploy the implant from the cartridge through at least a portion of the lumen such that the implant engages the eye tissue. The method can further include delivering a viscous material through the shaft.

[0011] In an interrelated implementation, described is a system for preparation of an implant from a patch of a material and ab interno insertion of the implant into an eye of a patient that includes a tissue cartridge having a nose cone and a distal shaft defining a lumen between the nose cone and a distal end region of the distal shaft; a cutting device configured to couple to the nose cone; and a delivery device configured to couple to the nose cone.

[0012] At least the distal end region of the distal shaft can be sized and shaped for insertion into an anterior chamber of the eye. The cutting device can include a base configured to receive the patch. The cutting device can include a cutting member configured to cut the patch of a material into the implant. The cutting device can further include a compacting tool configured to urge the implant into the lumen of the distal shaft. The delivery device can include an actuator configured to deploy the implant compacted within the lumen of the distal shaft into the eye.

[0013] In an interrelated implementation, described is a method of preparing an implant from a patch of a material for implantation into, and of inserting said implant into, an eye of a patient. The method includes coupling a tissue cartridge with a cutting device, the tissue cartridge having a shaft extending from a distal end of the tissue cartridge, at least a distal end region of the shaft sized and shaped for insertion into an anterior chamber of the eye. The shaft includes a lumen and the cutting device has a cutting member configured to cut the patch of a material. The method further includes cutting the patch with the cutting member to form the implant from the patch; compacting the implant within the lumen of the shaft; decoupling the tissue cartridge from the cutting device;

coupling the tissue cartridge to a delivery device; inserting the distal end region of the shaft into the anterior chamber of the eye; positioning the distal end region adjacent eye tissue; and actuating the delivery device to deploy the implant from the lumen such that the implant engages the eye tissue. The method can further include delivering a viscous material through the shaft.

[0014] In an interrelated implementation, described is a system for preparation of an implant and ab interno insertion of the implant into an eye of a patient including a tissue cartridge; and a delivery device. The tissue cartridge can include a shaft extending from a distal end of the tissue cartridge, at least a distal end region of the shaft sized and shaped for insertion into an anterior chamber of the eye. The shaft can include a lumen. The system can further include a cutting device having a cutting member configured to cut the patch of a material. Cutting the patch of a material with the cutting member can form an implant from the patch that is configured for implantation into the eye of the patient. The delivery device can include an actuator configured to deploy the implant positioned within the shaft through the lumen of the shaft into the eye. The tissue cartridge can include a nose cone assembly having the distal end region of the tissue cartridge and the shaft. The nose cone assembly can be reversibly coupled to the tissue cartridge and reversibly coupled to the delivery device. The shaft of the tissue cartridge can be configured to deliver a viscous material.

[0015] In an interrelated implementation, described is a method of preparing an implant for implantation into, and of inserting said implant into, an eye of a patient including cutting a patch of a material with a cutting member of a cutting device to form an implant from the patch; compacting the implant within a lumen of a shaft extending from a distal end of a tissue cartridge; decoupling at least a portion of the tissue cartridge from the cutting device; coupling the at least a portion of the tissue cartridge to a delivery device; inserting a distal end region of the shaft into the anterior chamber of the eye; positioning the distal end region adjacent eye tissue; and actuating the delivery device to deploy the implant from the tissue cartridge through at least a portion of the lumen such that the implant engages the eye tissue. The method can further include delivering a viscous material through the shaft.

[0016] In an interrelated implementation, described is a method of treating an eye with minimally-modified biological tissue. The biological tissue can be scleral tissue. Minimally-modifying the scleral tissue can include compressing the scleral tissue from a first size into a second, smaller size within a distal shaft. The distal shaft can be sized and shaped to be inserted through a self-sealing incision in a cornea of the eye into the anterior chamber. The method can further include deploying the compressed scleral tissue from the distal shaft between tissue layers near the iridocorneal angle. The compressed scleral tissue deployed from the distal shaft can return towards the first size. The method can further include treating glaucoma with the compressed scleral tissue.

[0017] In some variations, one or more of the following can optionally be included in any feasible combination in the above methods, apparatus, devices, and systems. More details are set forth in the accompanying drawings and the description below. Other features and advantages will be apparent from the description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] These and other aspects will now be described in detail with reference to the following drawings. Generally, the figures are not to scale in absolute terms or comparatively, but are intended to be illustrative. Also, relative placement of features and elements may be modified for the purpose of illustrative clarity.

[0019] FIGS. 1A-1B are cross-sectional views of a human eye showing the anterior and vitreous chambers of the eye with a stent being positioned in the eye in an example location;

[0020] FIG. 2 is a perspective view of a system according to an implementation;

[0021] FIGS. 3A and 3B illustrate an implementation of a tissue cartridge having a cover removed;

[0022] FIG. 3C illustrates the tissue cartridge with the cover installed;

[0023] FIG. 4A illustrates an implementation of a cutting device having a tissue cartridge installed and the cutter in the open configuration;

[0024] FIG. 4B illustrates the cutting device with the tissue cartridge installed and the cutter in the cut configuration;

[0025] FIG. 4C is a partial view of the cutting device of FIG. 4B showing the cutter;

[0026] FIG. 4D is a cross-sectional partial view of the cutting device of FIG. 4A;

[0027] FIG. 4E is a cross-sectional, partial view of the cutting device of FIG. 4B;

[0028] FIGS. 4F-4G illustrate the pusher in advanced and withdrawn configurations relative to the base of the cutting device;

[0029] FIG. 4H is a cross-sectional view of the cutting device of FIG. 4G;

[0030] FIGS. 4I-4J are cross-sectional partial view of the cutting device of FIG. 4F;

[0031] FIG. 5A illustrates an implementation of a delivery device having a tissue cartridge installed and the pusher in the advanced configuration;

[0032] FIG. 5B illustrates the delivery device of FIG. 5A with the cartridge withdrawn relative to the pusher;

[0033] FIG. 5C illustrates the tissue cartridge and distal end region of the delivery device;

[0034] FIG. 5D illustrates the tissue cartridge installed within the delivery device of FIG. 5C;

[0035] FIG. 5E illustrates the pusher of the delivery device advanced to deployment position;

[0036] FIG. 5F illustrates the tissue cartridge retracted by the delivery device to deploy the cut stent within the eye;

[0037] FIG. 6 is a perspective view of a system according to an interrelated implementation;

[0038] FIGS. 7A and 7B illustrate a tissue cartridge having a cover in a loading configuration;

[0039] FIG. 7C illustrates the tissue cartridge of FIGS. 7A-7B with the cover installed;

[0040] FIG. 8 illustrates an implementation of a cutting device and a tissue cartridge;

[0041] FIG. 9A illustrates an implementation of the cutting device having the tissue cartridge installed, the cutter in the cut configuration, and a nose cone of the tissue cartridge detached;

[0042] FIG. 9B illustrates an implementation of a delivery device having the nose cone of the tissue cartridge engaged and the pusher in the retracted configuration;

[0043] FIG. 9C illustrates the delivery device of FIG. 9B with the pusher advanced to the primed configuration;

[0044] FIG. 9D illustrates the delivery device of FIG. 9C with the nose cone retracted relative to the pusher;

[0045] FIG. 10A illustrates the nose cone prior to engagement with a distal end region of the delivery device;

[0046] FIG. 10B illustrates the nose cone after engagement with the distal end region of the delivery device and prior to attachment;

[0047] FIG. 10C illustrates the nose cone engaged and attached with the distal end region of the delivery device;

[0048] FIG. 11A illustrates the pusher in the first, retracted position;

[0049] FIG. 11B illustrates the pusher advanced to the second, primed position;

[0050] FIG. 11C shows the distal shaft positioned within the eye and the third actuator ready to be activated;

[0051] FIGS. 12A-12B are cross-sectional views of the device showing the first, retracted position of FIG. 11A;

[0052] FIGS. 12C-12D are cross-sectional views of the device showing the second, primed position of FIG. 11B;

[0053] FIGS. 13A-13B illustrate a reset mechanism of the delivery device;

[0054] FIGS. 14A-14H illustrate implementations of a cutting assembly for cutting and transferring

a stent to a portion of the tissue cartridge;

[0055] FIG. 14I illustrates in schematic an implementation of a nose cone assembly coupled to a cutting assembly;

[0056] FIGS. 15A-15B illustrate another implementation of a cutting device for cutting a stent.

[0057] It should be appreciated that the drawings are for example only and are not meant to be to scale. It is to be understood that devices described herein may include features not necessarily depicted in each figure.

DETAILED DESCRIPTION

[0058] Disclosed are implants, systems, and methods for increasing aqueous outflow from the anterior chamber of an eye. As will be described in detail below, ab interno outflow stenting using biological, cell-based or tissue-based materials provides biocompatible aqueous outflow enhancement with improved tolerability and safety over conventional shunts. In an example implementation, a biologic tissue or biologically-derived material is harvested or generated in vitro and formed into an implant, also referred to herein as a stent, using a cutting device, also referred to herein as a trephining device or cutting tool. In an implementation, the stent is an elongated body or material that has an internal lumen to provide a pathway for drainage. In a preferred implementation, the stent is an elongated body or strip of tissue that does not have an internal lumen and is configured to maintain the cleft and provide supraciliary stenting. Lumen-based devices can be limited by the lumen acting as a tract for fibrotic occlusion. The stent formed from the tissue is then implanted into the eye via an ab interno delivery pathway to provide aqueous outflow from the anterior chamber. The stents described herein can be used as a phacoemulsification adjunct or stand-alone treatment to glaucoma as a micro-invasive glaucoma surgery (MIGS) treatment.

[0059] Use of the terms like stent, implant, shunt, bio-tissue, or tissue is not intended to be limiting to any one structure or material. The structure implanted can, but need not be a material that is absorbed substantially into the eye tissue after placement in the eye such that, once absorbed, a space may remain where the structure was previously located. The structure once implanted may also remain in place for an extended period and not substantially erode or absorb.

[0060] As will be described in more detail below, the stents described herein can be made from biologically-derived material that does not cause toxic or injurious effects once implanted in a patient.

[0061] The term “biologically-derived material” includes naturally-occurring biological materials and synthesized biological materials and combinations thereof that are suitable for implantation into the eye. Biologically-derived material includes a material that is a natural biostructure having a biological arrangement naturally found within a mammalian subject including organs or parts of organs formed of tissues, and tissues formed of materials grouped together according to structure and function. Biologically-derived material includes tissues such as corneal, scleral, or cartilaginous tissues. Tissues considered herein can include any of a variety of tissues including muscle, epithelial, connective, and nervous tissues. Biologically-derived material includes tissue harvested from a donor or the patient, organs, parts of organs, and tissues from a subject including a piece of tissue suitable for transplant including an autograft, allograft, and xenograft material. Biologically-derived material includes naturally-occurring biological material including any material naturally found in the body of a mammal. Biologically-derived material as used herein also includes material that is engineered to have a biological arrangement similar to a natural biostructure. For example, the material can be synthesized using in vitro techniques such as by seeding a three-dimensional scaffold or matrix with appropriate cells, engineered or 3D printing material to form a bio-construct suitable for implantation. Biologically-derived material as used herein also includes material that is cell-derived including stem cell(s)-derived material. In some implementations, the biologically-derived material includes an injectable hyaluronate hydrogels or viscomaterials such as GEL-ONE Cross-linked Hyaluronate (Zimmer).

[0062] In some implementations, the biostent may be an engineered or 3D printed material formed in the shape of a tube with a lumen extending from a proximal opening to a distal opening. The tube may also be printed to incorporate a plurality of openings throughout. For example, a wall of the printed material can be designed to have a plurality of openings such that a liquid within the lumen can seep or flow outward through the wall of the tube such that the tube is sufficiently porous to ensure drainage of aqueous from the eye. The tube may be printed to have a dimension that is modified at or near the time of delivery. For example, a 3D printed material may be engineered to have a first dimension that is convenient for manipulating manually. At or near the time of delivery, the 3D printed material may be cut to a size more suitable for implantation in the eye. Where a patch of material is described as being cut or trephined into a stent prior to implantation it should be appreciated that the patch of material can be a printed material having a particular 3-dimensional shape (e.g., including tubular) and is cut into a stent by cutting to a shorter, desired length. Thus, in certain implementations, the stents described herein need not be solid and can also incorporate a lumen.

[0063] The biologically-derived material, sometimes referred to herein as bio-tissue or bio-material, that is used to form the stent can vary and can be, for example, corneal tissue, scleral tissue, cartilaginous tissue, collagenous tissue, or other firm biologic tissue. The bio-tissue can be of hydrophilic or hydrophobic nature. The bio-tissue can include or be impregnated with one or more therapeutic agents for additional treatment of an eye disease process.

[0064] The bio-stent material can be used in combination with one or more therapeutic agents such that it can be used to additionally deliver the agent to the eye. In an implementation, the bio-tissue can be embedded with slow-release pellets or soaked in a therapeutic agent for slow-release delivery to the target tissue.

[0065] Non-biologic material includes synthetic materials prepared through artificial synthesis, processing, or manufacture that may be biologically compatible, but that are not cell-based or tissue-based. For example, non-biologic material includes polymers, copolymers, polymer blends, and plastics. Non-biologic material includes inorganic polymers such as silicone rubber, polysiloxanes, polysilanes, and organic polymers such as polyethylene, polypropylene, polyvinyls, polyimide, etc.

[0066] Regardless the source or type of biologically-derived material, the material can be cut or trephined into an elongated shape suitable for stenting and implantation in the eye. This cutting process of the tissue can be performed before the surgical implantation process or during the surgical implantation process. The stent(s) implanted in the eye may have a structure and/or permeability that allows for aqueous outflow from the anterior chamber when positioned within a cyclodialysis cleft.

[0067] The biologically-derived material can be minimally modified or minimally manipulated tissue for use in the eye. The minimally modified biologically-derived material does not involve the combination of the material with another article, except, for example, water, sterilizing, preserving, cryopreservatives, storage agent, and/or pharmaceutical or therapeutic agent(s), and the like. The minimally modified biologically-derived material does not have a systemic effect once implanted and is not dependent upon the metabolic activity of any living cells for its primary function. The biologically-derived material can be minimally manipulated during each step of the method of preparation and use so that the original relevant characteristics of the biologic tissue is maintained. The cut stent can be a structural tissue that physically supports or serves as a barrier or conduit, for example, by maintaining at least in part a ciliary cleft formed in the eye. The stent cut from the biologically-derived material can be minimally manipulated such as by compressing, compacting, folding, rolling, or other sort of temporary manipulation of the cut stent that once freed from the forces applying the compression or compaction allows for the material to return towards its original structure. Thus, the minimal manipulation can mechanically change the size or shape of the cut tissue temporarily while still maintaining the original relevant characteristics of the tissue relating

to its utility for reconstruction, repair, or replacement once freed from that mechanical change. As an example, the biologically-derived material can be sclera that is cut into a shape that is oversized in relation to an inner diameter of a delivery tube through which the stent is implanted. The minimal manipulation of the cut stent can include temporarily compacting the scleral material into a lumen of the delivery shaft such that after implantation in the eye, the cut stent tends to return towards its original cut size. Although the biologically-derived material is described herein in the context of being cut into a stent like implant that can maintain a cleft for outflow of aqueous, other methods are considered herein. For example, the biologically-derived material can be compressed into a plug that is then implanted in a region of the eye for another purpose such as stenting, occlusion of traumatic ruptures, over-filtering bleb, posterior wall rupture, and other indications. [0068] FIGS. 1A-1B are cross-sectional views of a human eye showing the anterior chamber AC and vitreous chamber VC of the eye. A stent **105** can be positioned inside the eye in an implanted location such that at least a first portion of the stent **105** is positioned in the anterior chamber AC and a second portion of the stent **105** is positioned within tissues such as within the supraciliary space and/or suprachoroidal space of the eye. The stent **105** is sized and shaped such that the stent **105** can be positioned in such a configuration. The stent **105** provides or otherwise serves as a passageway for the flow of aqueous humor away from the anterior chamber AC (e.g. to the supraciliary space and/or suprachoroidal space). In FIGS. 1A-1B, the stent **105** is represented schematically as an elongated body relative to a delivery shaft **210**. It should be appreciated that the size and shape of the stent **105** can vary. Additionally, the size and shape of the stent **105** prior to insertion within the delivery shaft **210** can change upon insertion into the delivery shaft **210** and can change after deployment from the delivery shaft **210**.

[0069] The stent **105** can be implanted ab interno, for example, through a clear corneal incision or a scleral incision. The stent can be implanted to create an opening or cleft for augmented outflow communication between the anterior chamber AC and the supraciliary space, the anterior chamber AC and the suprachoroidal space, the anterior chamber AC and Schlemm's Canal, or the anterior chamber AC and the sub-conjunctival space, or any other ocular compartment, tissue or interface where trans-scleral, sub-scleral, or supra-scleral occlusion, stenting, and/or tissue reinforcing are clinically indicated. In a preferred implementation, the stent **105** is implanted such that a distal end is positioned within a supraciliary position and the proximal end is positioned within the anterior chamber AC to provide a supraciliary cleft. The distal end of the stent **105** can be positioned between other anatomical parts of the eye.

[0070] Conventional glaucoma stenting devices are typically formed of non-biological materials such as polyimide or other synthetic materials that can cause endothelial tissue damage leading to progressive, long-term, and irreversible corneal endothelial loss. The stent materials described herein can reduce and/or eliminate these risks of tissue damage while still providing enhanced aqueous humor outflow.

[0071] The stent **105** described herein can be formed of any of a variety of biologically-derived materials having a permeability and/or structure that allows for aqueous filtration therethrough. The stent **105** can be formed of a biologically-derived material that is harvested, engineered, grown, or otherwise manufactured. The biologically-derived stent material can be obtained or harvested from a patient or from donors. The biologically-derived stent material can be harvested before or during surgery. The biologically-derived stent material can be synthetic bio-tissue created using in vitro techniques. The biologically-derived material can be stem cell generated or bioengineered. The tissue can be generated via in situ cellular or non-cellular growth. In an example implementation, the tissue can be 3D printed during manufacture. The biologically-derived material can be minimally manipulated material and retain its original structural characteristic as a tissue.

[0072] The 3D printed tissue can be printed as a larger patch of material that is then cut at the time of surgery as described elsewhere herein. Alternatively, the 3D printed tissue can be printed to have the dimensions of the final implantable stent. In this implementation, the 3D printed material need

not be cut before implantation, but can be implanted directly. For example, the 3D printed stent can be printed directly into a cartridge that is configured to operatively couple with the delivery device described herein, which is in turn used to deploy the 3D printed stent into the eye. The 3D printed stent can be generated using the 3D printing process described in *Biofabrication*, 2019; 11 (3).

[0073] In an example implementation, the stent **105** is made of a bio-tissue. The biologically-derived material can be corneal tissue and/or non-corneal tissue. The biologically-derived material may include corneal, scleral, collagenous or cartilaginous tissue. In an implementation, the biologically-derived stent material can be denuded corneal stromal tissue without epithelium and endothelium that is porous and has hydrophilic permeability to allow aqueous filtration. The biologically-derived material can be minimally manipulated sclera that retains its original structural characteristic as a tissue. The biologically-derived material of the stent **105** can, but need not be incorporated into the eye's inherent anatomy after placement in the eye. The stent can cause the surrounding tissue to form a pathway that remains open for an extended period, even after absorption of the stent. The biologically-derived stent material may not significantly absorb or be incorporated into the eye's anatomy such that the stent **105** remains implanted for an extended period of time or indefinitely, as needed.

[0074] In other implementations, the stent **105** material may be manufactured of a complex carbohydrate or a collagen that is non-inflammatory. The stent **105** may also be formed of a biodegradable or bioabsorbable material including biodegradable polymers including hydroxyaliphatic carboxylic acids, either homo- or copolymers, such as polylactic acid, polyglycolic acid, polylactic glycolic acid; polysaccharides such as cellulose or cellulose derivatives such as ethyl cellulose, cross-linked or uncross-linked sodium carboxymethyl cellulose, sodium carboxymethylcellulose starch, cellulose ethers, cellulose esters such as cellulose acetate, cellulose acetate phthallate, hydroxypropylmethyl cellulose phthallate and calcium alginate, polypropylene, polybutyrates, polycarbonate, acrylate polymers such as polymethacrylates, polyanhydrides, polyvalerates, polycaprolactones such as poly-ε-caprolactone, polydimethylsiloxane, polyamides, polyvinylpyrrolidone, polyvinylalcohol phthallate, waxes such as paraffin wax and white beeswax, natural oils, shellac, zein, or a mixture. The stent **105** may be formed of hyaluronate hydrogels or viscomaterials.

[0075] As mentioned, the biologically-derived stent material can have a permeability or porosity that allows for aqueous filtration for sufficient control or regulation of intraocular pressure. Permeable bio-tissues described herein (e.g. sclera, cornea, collagen, etc.) are preferred stent materials, however, any bio-tissue, even if impermeable, is considered herein as a potential stent material to serve as a structural spacer that keeps the cyclodialysis open. Preferably, the material of the stent can create a gap that allows fluid to flow. The gap created can run longitudinally along each side of the stent. If the material of the stent is permeable, more fluid can pass through the cyclodialysis than if the stent material is impermeable and the fluid is required to pass along the outside of the stent. Thus, the material considered herein need not be porous in order to provide the desired function, however, the function can be enhanced by the porosity of the material.

[0076] Generally, the biologically-derived stent material has some firmness and intraocular durability such that it can maintain outflow from the anterior chamber, however, is less stiff than conventional non-biologically-derived polyimide shunts used in the treatment of glaucoma (e.g. CYPASS, Alcon). The stent material may have a sufficient structure to serve as a spacer to prop open a sustained supraciliary outflow. The stent material can maintain its structural height or thickness once implanted within the cyclodialysis such that fluid flow through or around the stent is provided. In some implementations, the cut stent is minimally manipulated by compressing or compacting into a delivery shaft so that the size and/or shape of the cut stent is reduced from a first size into a second, smaller size within the shaft. The delivery shaft can be sized and shaped to be inserted through a cornea (such as a self-sealing incision in a cornea) into the anterior chamber and advanced towards the iridocorneal angle. The delivery shaft can deploy the compacted stent

between tissue layers near the angle. Once the compacted stent is deployed from the delivery shaft it can begin to return towards its original shape and/or size. The cut stent, once implanted, can take on a shape and/or size that is smaller from its original shape and/or size or that is the same as its original shape and/or size. The minimally-modified biological tissue can be used to treat glaucoma. Biologically-derived stent material provides advantages in terms of biocompatibility, anatomic conformity, and aqueous permeability compared to conventional non-biological materials such as polyimide. Biologically-derived stent material can provide better conformability and compliance to the scleral wall and can be less likely to cause endothelial and scleral erosion/loss over time and with chronic eye rubbing and blinking.

[0077] Typically, allograft tissue for implantation into the eye is handled delicately so as not to modify it from its original state. The cut stents described herein need not be handled so delicately and instead can be minimally-modified by compressing or compacting or otherwise wedging into a smaller space for ab-interno delivery into the eye for intraocular stenting, occlusion, reinforcement through a corneal or scleral incision or puncture (less than about 3.5 mm).

[0078] In an implementation, the material used to form the stent is provided as an uncut patch of material configured to be manually loaded within a cartridge **200**. The uncut patch of material can also be cut by a cutting assembly that is independent of a cartridge **200** and then transferred into a region of a cartridge **200**. As will be discussed in greater detail below, the cutting can be done at the time of surgery or prior to surgery. In certain implementations, the stent is formed by 3D printing and can be printed into a desired final dimension for the stent or can be printed as a patch of material that is then cut at the time of or prior to surgery. The cutting achieved by the devices described herein can provide thin strips of material that can be implanted in the eye to provide regulation of aqueous outflow. The process of cutting or trephining can position the cut implant within a conduit or lumen of the cartridge such that the cut implant held within the cartridge may be subsequently delivered from the delivery device without needing to remove or transfer the cut implant from the cartridge. Alternatively, the cutting can be performed independently of transferring the cut implant into a delivery device. The cutting and transferring of the cut implant into a delivery device can be independent steps performed by independent tools or assemblies. For example, the system can incorporate a first device that is used for cutting the patch of material into a cut implant, a second device used to transfer the cut implant into a delivery device, and a third device used to deploy the cut implant from the delivery device into the eye. It should be appreciated that the cutting, transferring, and deploying can be integrated into a single device or one or more can be independent devices used in conjunction with one another to transition a patch of material into a cut implant for deployment in an eye. In a preferred embodiment, the cutting and transferring of the cut implant are integrated into a first device and the deployment of the cut implant in an eye is in a second device.

[0079] The term “patch of material” as used herein refers to a piece of biologically-derived material having a size along at least one dimension that is greater than a size of the stent cut from the patch of material and implanted in the subject. In some implementations, the patch of material can have a generally square shape and the stent cut or trephined from the patch of material can have a generally rectangular shape. For example, the patch of material can be about 7 mm wide×7 mm long×0.55 mm thick and the stent cut from the patch of material can be 0.3-1.0 mm wide×7 mm long×0.55 mm thick. The dimensions of the patch of material and the cut stent can vary. The patch of material prior to cutting can be between about 5 mm to about 10 mm wide, between about 5 mm up to about 10 mm long, and between about 0.25 mm to about 2 mm thick. The stent cut from the patch of material can be between about 0.3 mm up to about 2 mm wide, preferably between 0.7 mm to 1.0 mm wide. The stent cut from the patch of material can be between about 5 mm up to about 10 mm long. The stent cut from the patch of material can be between 0.25 mm to about 2 mm thick. The patch of material and the cut stent can each have the same length and the same thickness, but differ from one another in width. The patch of material and the stent cut from the patch of

material can also have different lengths and thicknesses. For example, the patch of material can have a first thickness and the stent cut from the patch of material have the same thickness, but when implanted can be folded or rolled into a different thickness from the patch of material. The cut stent need not be rectangular in shape and can have a non-rectangular shape such as an angular wedge or any of a variety of shapes to provide a particular clinical result. For example, a stent cut to the shape of a “dog bone” having enlarged distal and proximal ends may provide additional fixation within the target tissues. The stent can be cut to have a narrow elongate shape on a leading end and an enlarged dimension on a trailing end to provide ease of insertion as well as at least one end providing fixation.

[0080] In some implementations, the patch of material can be a relatively larger width (e.g., 10 mm×10 mm) and the stent cut from the patch to a strip having a much smaller width (e.g., about 1.0 mm to about 1.5 mm) and the cut stent then compacted into a delivery conduit having an inner diameter of about 0.8 mm so that the width of the stent substantially fills the inner diameter. A stent can substantially fill the inner diameter of the delivery conduit even if the stent is not oversized relative to that conduit and thus, remains uncompacted. The stent can be oversized relative to the inner dimension of the conduit and be compacted into the conduit to substantially fill it.

Additionally, the dimension of the cut stent can vary depending on the dimension of the conduit the stent is to be deployed through. For example, the inner diameter of the delivery conduit can be about 600 microns to about 800 microns. Thus, the stent can be cut or trephined to any of a variety of sizes depending on whether or not the stent is to be compacted into the delivery conduit and depending upon the inner dimension of that delivery conduit.

[0081] The stent cut from the patch of material can have a width, a length, and a thickness. In an implementation, the width of the stent cut from the patch of material using the cutting devices described herein can be at least 100 microns up to about 1500 microns, or between 100 microns up to 1200 microns, or between 100 microns and 900 microns, or between 300 microns and 600 microns. The stent cut from a patch of material can have a width of at least about 100 microns and a width of no more than 1500 microns, 1400 microns, 1300 microns, 1200 microns, 1100 microns, 1000 microns, 900 microns, no more than 800 microns, no more than 700 microns, no more than 600 microns, no more than 500 microns, no more than 400 microns, no more than 300 microns, or no more than 200 microns. The length of the stent cut from a patch of material can vary depending on the location of stent implantation. In some implementations, the stent has a length that is between 1 mm and 10 mm, or more preferably between 3 mm and 8 mm long. The thickness of the stent cut from the patch of material can be from 100 microns up to about 800 microns, or from 150 microns up to about 600 microns. In an implementation, the biological material forming the stent can have a thickness that is no smaller than 100 microns and no larger than 5 mm. The thickness of the stent can also depend on whether the stent is folded or rolled upon implantation such that a patch of material having a thickness of just 250 microns can cut into a stent and the stent folded at implantation to double the thickness to about 500 microns. The thickness of the stent can also depend upon what biologically-derived material is used. For example, scleral tissue or corneal tissue can often have a thickness of around 400 microns, but following harvest can shrink to about 250-300 microns. As such, a stent cut from a shrunken patch of corneal tissue may have a thickness of just 250 microns.

[0082] In some implementations, which is described in more detail below, the stent cut from the patch of material is cut so as to substantially fill the conduit through which it is advanced for delivery. In other implementations, the stent can be cut into an implant that is oversized relative to a dimension of a conduit through which it is deployed. In this implementation, the stent can be cut to have a first size, which is oversized compared to the inner dimension of the delivery conduit. The oversized stent can be primed within the delivery conduit such as by compacting or compressing with a tool so that the stent when primed within the conduit takes on a second, smaller size. Upon deployment in the eye and release of the stent from the delivery conduit, the stent may

achieve a third size approaching its original first size. This will be described in more detail below.

[0083] In a non-limiting example, bio-tissue stent has dimensions no smaller than 0.1 mm and no larger than 8 mm in any direction and a thickness of not smaller than 50 microns and not larger than 8 mm. In a non-limiting example, the stent is about 6 mm in length by 300-600 microns wide by 150-600 microns thick. The cutting can be no smaller than 1 mm and no larger than 8 mm in any direction. In a non-limiting example, the cut tissue has dimensions of 100-800 microns in width and 1 mm-10 mm in length. It should be appreciated that multiple stents may be delivered to one or more target locations during an implantation procedure.

[0084] FIGS. 2 and 6 show interrelated implementations of a system **100** for preparation and delivery of a biologic intraocular stent for increasing aqueous outflow and lowering of intraocular pressure. The system **100** can include a tissue cartridge **200** having at least a portion configured to be reversibly and operatively coupled with a cutting device **300** and a delivery device **400**.

[0085] Each of the systems **100** can be provided without a cutting device **300** and include only the tissue cartridge **200** and the delivery device **400**. In this implementation, the tissue cartridge **200** can include a pre-cut stent **105** within the cartridge **200** that is ready to be engaged with the delivery device **400** for deployment into the eye. The cartridge **200** with the pre-cut stent **105** can be immersed within a stable solution. Thus, where the systems are described as including a cutting device **300**, it should be appreciated that the cutting device **300** may not be used at the time of surgery and instead the stent **105** provided in a pre-cut and/or pre-primed configuration within at least a portion of the delivery device **400** or the tissue cartridge **200**.

[0086] FIG. 2 shows a first cartridge **200** shown separated from the cutting device **300** and another cartridge **200** installed with the delivery device. The cartridge **200** is configured to receive a patch of material **101** within the cartridge **200** and fix the patch of material **101** in preparation for cutting by the cutting device **300**. The cutting device **300** when operatively engaged with the cartridge **200** is configured to form the biologic intraocular stent **105** from the patch of material **101** held within the cartridge **200**. The delivery device **400** when operatively engaged with the cartridge **200** is configured to deliver the cut implant **105** from the cartridge **200** to the implanted location. The tissue cartridge **200** in the implementation of FIG. 2 is configured to mate with both the cutting device **300** and the delivery device **400** such that the entire tissue cartridge **200** is removed from and transferred between the two devices **300**, **400** of the system **100**.

[0087] FIG. 6 shows an interrelated implementation of the system **100** and includes a tissue cartridge **200** configured to be operatively coupled with a cutting device **300** and the delivery device **400**. However, the entire tissue cartridge **200** need not be fully removed from the cutting device **300** in order to couple with the delivery device **400**. In this implementation, the tissue cartridge **200** can include a distal nose cone assembly **274** that is configured to uncouple from a proximal portion **207** of the cartridge **200** and couple with the delivery device **400**. The nose cone assembly **274** can include at least a portion of the distal portion **205** such as a nose cone **275** and the shaft **210** extending distally from the nose cone **275**.

[0088] In still further implementations, the cartridge **200** need not include a portion configured to receive a patch of material **101** within the cartridge **200**. For example, the cartridge **200** can include only a nose cone assembly **274** including a nose cone **275** having a distal shaft **210**. The nose cone **275** with the distal shaft **210** can be coupled to a cutting device **300** that is configured to receive the patch of material **101** within at least a region and fix the patch of material **101** in preparation for cutting by the cutting device **300**. The nose cone **275** and distal shaft **210** can be arranged relative to the cutting device **300** so that the cut stent can be transferred into it for deployment in the eye. FIG. 14I illustrates in schematic a nose cone assembly **274** coupled to a cutting assembly **500**. The nose cone assembly **274** includes a nose cone **275** having a proximal end coupled to the cutting assembly **500** and a distal shaft **210** extending out from the nose cone **275** along longitudinal axis A. The cutting assembly **500** can be part of a cutting device **300** as described herein.

[0089] A cartridge can include any of a variety of structural arrangements as described herein, but

generally refers to a component that is transferrable between two or more devices. The cartridge can be transferrable between a cutting device and a delivery device. The cartridge can be configured to hold a patch of material for cutting into a stent as well as provide a conduit for deploying the stent into the eye. The cartridge need not be configured to hold the patch of material for cutting, however. The cartridge can include the shaft configured to receive the cut stent from the cutting assembly to then deploy the stent into the eye from the shaft. Any of a variety of configurations are described and considered herein.

[0090] Each of these systems and their respective components will be described in more detail herein.

[0091] FIG. 2 and also FIGS. 3A-3C show an implementation of the tissue cartridge **200** configured to hold the patch of material for cutting and for providing a conduit for deploying the cut stent into the eye. The cartridge **200** can include a distal portion **205** coupled to and extending distally from a proximal portion **207**. The distal portion **205** can include an elongate member or shaft **210** having an inner conduit or lumen **238** that is sized for containing and deploying the stent **105**. The proximal portion **207** can include a base **224** and a cover **214** movably attached to the base **224**. The proximal portion **207** is intended to remain outside the eye while the distal portion **205** is configured to insert within the eye to deploy the stent **105** within the target tissues. The implant **105** can be advanced from the proximal portion **207** of the cartridge **200** into a deployment positioned within the distal portion **205** of the cartridge **200**. The distal portion **205** of the cartridge **200** is insertable into the anterior chamber of the eye so that it may be positioned adjacent eye tissue within which the implant **105** is deployed from the cartridge **200** into the eye tissue. For example, the distal portion **205** of the cartridge **200** can be inserted ab interno into the anterior chamber through a corneal incision, while the proximal portion **207** of the cartridge **200** remains outside the eye (e.g., coupled to the delivery instrument **400**).

[0092] FIG. 6 and also FIGS. 7A-7C illustrate another implementation of a tissue cartridge **200** configured to hold the patch of material for cutting and for providing a conduit for deploying the cut stent into the eye. The tissue cartridge **200** can include a distal portion **205** coupled to and extending distally from a proximal portion **207** that includes a shaft **210** having an inner conduit or lumen **238** (visible in FIG. 14I) sized for containing and deploying the stent **105**. The proximal portion **207** can also include a base **224** and a cover **214** movably attached to the base **224**. The distal portion **205** and shaft **210** can be removably attached to the proximal portion **207** of the cartridge **200**. For example, the proximal portion **207** can remain within the cutting device **300** and a removable nose cone assembly **274** comprising the nose cone **275** and the shaft **210** can be disengaged from the proximal portion **207** and engaged with the delivery instrument **400** (see FIGS. 9A-9D).

[0093] It should be appreciated that the distal portion **205** of the cartridge **200** can be useful for other delivery pathways (e.g., trans-scleral delivery). Deploying the implant **105** into the eye tissue can include the implant **105** residing at least in part between a ciliary body and a sclera of the eye. The implant **105** can reside between the ciliary body and the sclera within a cyclodialysis cleft.

[0094] The shaft **210** of the cartridge **200** (also referred to herein as an introducer tube, applicator, conduit, or delivery body) extending in a distal direction outward from the proximal portion **207** of the cartridge **200** includes at least a portion that extends along a longitudinal axis A. At least another portion of the shaft **210** can be angled, curved, or flexible such that it can form a distal curve or a bend away from the longitudinal axis A. In some implementations, the shaft **210** can include a flexible portion and a rigid portion such that depending on relative position of the portions results in a change in shape of the shaft. The implementation shown in FIGS. 3A-3C and also FIGS. 7A-7C has a proximal portion that extends along the longitudinal axis A and a distal end region **212** that curves downward away from the longitudinal axis A. The distal end region **212** can include an opening **230** from the lumen **238** through which the stent **105** can be deployed. The opening **230** from the lumen **238** can be positioned within a plane that is perpendicular to a plane

of the longitudinal axis A of the distal end region **212** of the shaft **210**. The opening **230** from the lumen **238** can be positioned within a plane that is at an angle relative to the longitudinal axis A of the distal end region **212** of the shaft **210**. The distal end region **212** of the shaft **210** can be beveled such that the opening **230** into the lumen **238** is elongated rather than circular and a distal-most tip **216** of the shaft **210** extends beyond the opening **230**. The distal-most tip **216** of the shaft **210** can be a pointed tip or a blunt tip that is squared off such that it does not form a point. The shape of the opening **230** can be a function of the overall cross-section of the shaft **210** at the distal end region **212** as well as the angle of the opening **230** relative to the longitudinal axis A of the distal end region **212**. For example, if the distal end region **212** of the shaft **210** has a rectangular cross-section and the opening **230** is cut perpendicular relative to the longitudinal axis A, the opening **230** and the cross-sectional shape of the shaft **210** are substantially matched. If the shaft **210** has a rectangular cross-section and the opening **230** is cut less than perpendicular relative to the longitudinal axis A, the opening **230** may have an elongated rectangular shape compared to the rectangular shape of the shaft **210**. The opening **230** may also have a first shape near the heel of the bevel and a second shape near the distal-most tip **216**. For example, the opening **230** near the heel of the bevel may be rounded and the opening **230** near the distal-most tip **216** may be squared-off. It should also be appreciated that the opening **230** need not be at the distal-most end of the shaft **210**. The opening **230** can be formed in a sidewall of the shaft **210** such that the stent **210** is urged out of the lumen **238** along a direction that is angled relative to the longitudinal axis of the lumen **230**. The opening **230** can be positioned in the shaft **210** relative to the cartridge **200** such that it is positioned on a forward end, a lower side, an upper side, and/or another side of the shaft **210**. The distal end region **212** of the shaft **210** can have a cross-sectional shape that is circular, oval, rounded rectangle, rectangle, rounded square, square, diamond, tear drop, or other shape and the distal-most tip **216** have a tip shape that varies, including blunt tip, bullet tip, spatula tip, or pointed tip. The distal end region of the shaft **210** can have any of a variety of configurations known in the ophthalmic arts.

[0095] The shaft **210** can be used to create a cyclodialysis cleft within the supraciliary space. The distal end region of the shaft **210** can be shaped to form the cleft as well as provide a conduit for a material to be delivered into the supraciliary space of the eye. The shaft **210** can also be used to deliver a viscous material such as viscoelastic fluid or a non-viscous material such as the sclera tissue. For example, viscoelastic can be delivered to a region of the eye through the shaft **210** prior to, during, and/or after implantation of the stent. The corneal incision can be created with a scalpel or other tool and the shaft **210** inserted through the incision and the distal end of the shaft **210** navigated to a desired location for delivery. The distal end of the shaft **210** can include a spatula that can be used to separate tissue layers and create the cyclodialysis cleft in the supraciliary space between the sclera and ciliary body. The dimensions, surface finish, and shape of the distal end can minimize trauma. The shaft **210** can additionally include one or more markers providing user information regarding distance of insertion. A distal end region of the shaft **210** can include one or more markers for goniometric reference for how deeply the tongue of the shaft **210** has been inserted into the supraciliary space. The length of the shaft **210** is sufficient to allow the device to be used from a temporal or superior position.

[0096] The shaft **210** of the cartridge **200** has a size and shape configured for ab interno delivery through a clear corneal incision to permit passage of the stent **105** out the distal end of the shaft **210**. In at least some implementations, the distal end region **212** of the shaft **210** is sized to extend through an incision that is about 1 mm in length. In another implementation, the distal end region **212** of the shaft **210** is sized to extend through an incision that is no greater than about 2.5 mm in length. In another implementation, the distal end region **212** of the shaft **210** is sized to extend through an incision that is between 1.5 mm to 2.85 mm in length. In some implementations, the maximum outer diameter of the shaft **210** is no greater than 1.3 mm. The distal-most tip **216** of the shaft **210** can be blunt or sharp. A blunt distal-most tip **216** of the shaft **210** allows for dissecting

between tissues of the eye without penetrating or cutting the tissues for positioning the stent **105**. For example, the distal-most tip **216** of the shaft **210** can be configured to bluntly dissect between the ciliary body CB and the sclera S (e.g., the supraciliary space) while the stent **105** remains fully encased within the shaft **210** during the blunt dissection. In an alternative implementation, the distal-most tip **216** of the shaft **210** has a sharp cutting configuration for dissecting application and implantation through the scleral wall into the subconjunctival space. In yet another embodiment, the distal-most tip **216** can have a cutting configuration for dissecting and implantation into the Schlemm's Canal or trans-sclerally.

[0097] The shaft **210** can be a hypotube that is no greater than about 18 G (0.050" OD, 0.033" ID), 20 G (0.036" OD, 0.023" ID), 21 G (0.032" OD, 0.020" ID), 22 G (0.028" OD, 0.016" ID), 23 G (0.025" OD, 0.013" ID), 25 G (0.020" OD, 0.010" ID), 27 G (0.016" OD, 0.008" ID), 30 G (0.012" OD, 0.006" ID), or 32 G (0.009" OD, 0.004" ID). In some implementations, the shaft **210** is a hypotube having an inner diameter that is less than about 0.036" down to about 0.009". The system can incorporate a 600 micron shaft **210** or an 800 micron shaft **210**. Other sizes for the shaft **210** are considered herein depending on particular patient conditions and clinical needs.

[0098] In preferred implementations, the stents described herein can be formed as solid strips of material without any lumen although it should be appreciated the stent may have also include a lumen. Thus, the stents are generally not deliverable over a guidewire as many conventional glaucoma shunts are. Additionally, the stents described herein can be formed of relatively soft tissue that is more fragile as typical shunts, which are formed of more rigid polymeric or metal material. Rigid shunts can be implanted such that a distal end of the shunt is used to create a blunt dissection at the interface of the tissues through which the shunt is being inserted. The stents described herein are preferably deployed using a retractable sleeved type of injector or introducer tube that once in proper anatomic position can be retracted leaving the stent more gently externalized and precisely positioned.

[0099] The dimensions of the shaft **210** can be selected based on the dimensions desired for the stent to be implanted. The stents **105** can have a dimension that substantially fills the inner lumen **238** of the shaft **210** (or the inner lumen of at least a portion of the shaft **210** through which it is delivered) such that the stent may be urged distally through that portion. In some implementations, the stent substantially filling the lumen is urged distally without wrinkling or being damaged. In other implementations, the stent substantially filling the lumen is urged distally through the shaft **210** in a manner that compacts the tissue into a plug having a denser configuration than the stent when cut from the patch. The dimensional difference or gap between the width and height dimensions of the stent **105** and the inner dimensions of the conduit can be up to about 200% of the dimensions of the stent **105**. The maximum size of the conduit and the maximum size of the stent **105** are related. As an example, if the stent width is about 1 mm, the maximum dimension of the conduit can be 3 mm, which results in the total gap between the width of the stent and the outer wall of the conduit being 200% of the stent width. The gap may be less than 5-10% of the maximum dimension of the stent **105**. Generally, the smaller the gap between the stent **105** and the conduit, the better the result for advancing the stent **105** through the conduit. If the cross-sectional area of the shaft **210** is greater than 200% the cross-sectional area of the cut stent **105**, the stent **105** can buckle as it is being pushed through the shaft **210** to be implanted in the eye. The cross-sectional area of the shaft **210** and the cross-sectional area of the stent **105** are preferably substantially size-matched. The conduit can also be coated with a lubricious or low friction material (e.g., Teflon) to improve advancement of the stent **105** through the conduit during deployment.

[0100] The cross-sectional area of the shaft **210** can also be smaller than the cross-sectional area of the stent **105**. As mentioned above, the stent **105** can be cut to be oversized relative to the inner diameter of the shaft **210** so that the stent **105** is compressed, compacted, or otherwise minimally manipulated for delivery through the tube. The stent can be cut to have a first size, which is

oversized compared to the inner dimension of the shaft **210**. The oversized stent can be primed within the shaft such as by compacting with a compacting tool **420** so that the stent **105** when primed within the conduit takes on a second, smaller size. Upon deployment in the eye and release of the stent **105** from the shaft **210**, the stent **105** may achieve a third size approaching its original first size. Delivery and deployment will be described in more detail below.

[0101] The shaft **210** can, but need not be fully tubular, nor does the shaft **210** need to be circular in cross-section. For example, the shaft **210** can be circular, oval, square, rectangular, or other geometry in cross-section. Additionally, the entire length of the shaft **210** need not have the same cross-sectional shape or size. For example, a proximal end of the shaft **210** can have a first shape and a distal end of the shaft **210** can have a second shape. FIGS. 5A-5B shows the shaft **210** is rectangular in cross-section. The lumen **238** of the shaft **210** need not be a fully enclosed channel. For example, the shaft **210** may incorporate one or more fenestrations, openings, segmental windows, or walls having one or more discontinuities such that the lumen **238** through the shaft **210** is a partially enclosed channel.

[0102] Again with respect to FIGS. 3A-3C and also FIG. 7A-7C, the proximal portion **207** of the cartridge **200** can include a base **224**. A distal end region of the base **224** can be coupled to the shaft **210**. A proximal end region of the base **224** can include a recess **221** configured to receive the patch of material **101**. The recess **221** can include a projection **271** in the shape of an inverted V can project upward from a center line of the recess **221** that urges the centerline of the patch of material **101** upward while allowing the sides of the patch of material **101** to hang downward into corresponding channels **270** on either side of the centerline. FIGS. 7A-7C illustrate the proximal portion **207** of the cartridge **200** can be reversibly coupled to a nose cone assembly comprising the shaft **210** and the nose cone **274**.

[0103] The base **224** is configured to mate with the cover **214** and to at least partially enclose the recess **221** containing the patch of material **101**. The cover **214** is configured to engage at least some portion of the patch of material **101** to stabilize the tissue before and during cutting of the patch **101**, for example, with the cutting device **300**. In an implementation, the base **224** can include a slot **215** in an upper surface of the base **225** sized and shaped to receive the cover **214**. The cover **214** slides through the slot **215** until a lower surface of the cover **214** abuts against a receiver surface **218** of the base **224**. The contact between the lower surface of the cover **214** and the receiver surface **218** of the base **224** ensures the centerline of the patch of material **101** within the recess **221** is in contact with the lower surface of the cover **214** at the projection **271** (see FIG. 3C).

[0104] The cover **214** is shown in FIGS. 3A-3C as a completely removable element from the base **224**. The cover **214** and base **224** can optionally be coupled together by a hinge or other mechanical feature. For example, the cover **214** can rotate around a pivot axis of the hinge and stay connected to the base **224** even when in a configuration to reveal the recess **221**. FIGS. 7A-7C illustrate the cover **214** can toggle between an open and closed configuration by applying a downward pressure on a forward end of the cover **214** (FIG. 7A) to open the cover **214** and a downward pressure on a back end of the cover **214** to close the cover **214** (FIG. 7C). For example, the cover **214** can be lifted into an open configuration revealing the recess **221** of the base **224** within which the patch of material **101** can be positioned. When the cover **214** is positioned back into the closed configuration, the patch **101** can be compressed and/or tensioned between the cover **214** and the base **224**. The cartridge **200** can be inserted within **306** of the cutting device **300** once the cover is in the closed configuration (see FIG. 8)

[0105] The cover **214** (or some other element) can be configured to additionally apply an amount of tension on at least a portion of the patch of material **101**, such as stretching in an outward direction from the centerline of the patch of material **101** before cutting occurs as described in U.S. Pat. No. 10,695,218, issued Jun. 30, 2020, and is incorporated by reference herein in its entirety.

[0106] The patch of material **101** can be inserted by a user into the cartridge **200** at the time of

surgery. The patch of material **101** may be provided in a size that approximates the size of the recess **221** within the base **224**. The user may trim the patch of material **101** before installing it in the recess **221**. Alternatively, the cartridge **200** can be provided pre-loaded with a patch of material **101** positioned within the recess.

[0107] As mentioned elsewhere herein, the cartridge need not be configured to hold the patch of material **101** for cutting by the cutting device **300**. Rather, the cutting device **300** can be configured to hold the patch of material **101** for cutting and then transfer the cut stent into the cartridge that is coupled to the cutting device **300**. FIGS. **10A-10C** illustrate an implementation of a cartridge **200** that forms a nose cone **274** having a shaft **210** into which the cut stent can be loaded prior to insertion in the eye. The nose cone **274** can reversibly couple to a cutting device **300** and, once loaded with the cut stent, can be removed from the cutting device **300** and couple with the delivery device **400**. The cartridge **200** can be positioned relative to the cutting device **300** that is configured to hold the patch of material **101** and cut it into a stent **105**. The coupling between the cutting device **300** and the cartridge **200** can align the longitudinal axis of the distal shaft **210** relative to a region of the cutting device so that the cut stent **105** can be transferred into the distal shaft **210** such as with a rod or other tool that will be described in more detail below. The cartridge **200** with the distal shaft **210** having the stent **105** positioned inside it can then be uncoupled from the cutting device **300** and transferred to a portion of the delivery device **400**. Thus, the cartridge **200** need not include a portion configured to hold the patch of material **101** for cutting and instead includes a transferrable portion that can couple alternately with a region of the cutting device **300** and a region of the delivery device **400**. This will be described in more detail below.

[0108] FIGS. **4A-4J** and also FIG. **8** show implementations of a cutting device **300** having a cutting assembly for cutting a stent from a patch of material **101**. FIGS. **14A-14H** illustrate various implementations of a cutting assembly **500** that can be incorporated into the cutting device **300**. The cutting device **300** is configured to cut or otherwise prepare the biologically-derived tissue or patch of a material **101** having a first contour or shape (e.g., a wider, square sheet or patch of material) into a second contour or shape (e.g., a narrower, rectangular strip of material) that conforms to an implantable stent **105** having the dimensions described herein. The cutting performed using the cutting devices **300** described herein can involve guillotine, punch, rotating, sliding, rolling, or pivoting blade cutting motion. In some implementations, the cutting is performed orthogonal to the plane of the patch of material. In some implementations, the cutting is performed axially along the conduit of implantation such that the axis of cutting can be aligned, within, or parallel to an implantation conduit to allow unimpeded tissue loading and transfer for implantation without manipulating, tearing, or damaging the fragile stent tissue.

[0109] As mentioned above, the cutting process is preceded by a tissue fixation step wherein the biologically-derived tissue that forms the stent is firmly fixed between two appositional planar surfaces to ensure the tissue is not wrinkled or malformed and the subsequent cut is of accurate dimensions. The fixation can optionally provide compression as well as tension or stretching of the tissue within at least one plane to ensure clean cutting through the tissue. The cutting assembly **500** can hold the patch of material **101** prior to cutting or the patch of material **101** can be held within a region of the tissue cartridge **200** prior to cutting by the cutting assembly **500**. In some implementations, the cutting device **300** in combination with the cover **214** of the cartridge **200** can incorporate an anterior-to-posterior capture such that the material **101** to be cut is held fixed on the z-plane preventing movement prior to engaging the tissue with the cutting member **312**.

[0110] The cutting can be performed within a path or conduit formed within the cartridge **200**. Thus, implant **105** cut from the patch of material **101** can simultaneously or subsequently position the implant **105** within the conduit for delivery or align the implant **105** with the conduit for delivery so that the cut implant **105** can be delivered to the eye through the conduit without the cut implant **105** needing to be transferred from the cartridge **200**.

[0111] As an example, the patch of material **101** held within the recess **221** of the cartridge **200** is

cut by the cutting member **312** of the cutting device **300** forming a cut stent **105** within the recess **221** of the cartridge that can be urged distally from the recess **221** into the lumen **238** of the shaft **210** of the cartridge **200** so it can be deployed in the eye all without removing the cut stent **105** from the cartridge **200** or at least the distal portion **205** of the cartridge **200**.

[0112] With respect to FIGS. **4A-4B** and also FIG. **6**, the cutting device **300** can include a base **302** having a distal portion **305** and a proximal portion **307**. The distal portion **305** can include a distal opening or receptacle **306** sized and shaped to receive the proximal portion **207** of the cartridge **200**. The inner diameter of the receptacle **306** can be sufficient to receive an outer dimension of the proximal portion **207** so that the proximal portion **207** can be inserted a distance within the receptacle **306**. The cover **214** of the cartridge **200** positioned within the slot **215** to maintain the patch of material **101** within the recess **221**. An upper surface of the cover **214** can extend above the upper surface of the base **224** such that the outer dimension of the proximal portion **207** is keyed. In other words, the outer dimension of the cartridge **200** is keyed and can only be inserted within the receptacle **306** of the cutting device **300** in a single orientation (e.g., the cover **214** positioned on an upper side).

[0113] The cutting device **300** can additionally include a cutting assembly **500** having a cutting member **312** configured to cut the patch of material **101** within the recess **221** of the cartridge into a stent **105** (see FIG. **4C**). The configuration of the cutting member **312** can vary. In this configuration, the cutting member **312** can include at least a first blade **344a** and a second blade **344b** separated a distance from the first blade **344a**. The first and second blades **344a**, **344b** can be positioned above the patch of material **101** when the cartridge **200** is installed within the receptacle **306** of the cutting device **300**. Actuation of the cutting member **312** causes the first and second blades **344a**, **344b** to be urged towards the patch of material **101** cutting through the thickness thereby forming the stent **105**. The blades **344a**, **344b** can have a width along the longitudinal axis **A** of the cartridge **200** sufficient to cut a full length of the patch of material **101**. The distance between the blades **344a**, **344b** can be designed to achieve the width desired for the cut stent **105**.

[0114] In some implementations, the blades **344a**, **344b** can be positioned above the patch of material **101** to be cut and corresponding lower blades **345a**, **345b** can be positioned below the patch of material **101**. Thus, as the blades **344a**, **344b** are urged downward towards the patch of material **101**, they urge the patch of material **101** towards the lower blades **345a**, **345b** such that the corresponding upper and lower blades cut completely through the material **101** in two locations creating the stent **105**.

[0115] The cutting member **312** can be actuated by a user to move the blades. The cutting device **300** can include one or more handles **343** that movably coupled to the base **302** to actuate the cutting member **312**. The handle(s) **343** can be coupled by a hinge **317** such that the handles **343** rotate around a pivot axis **P** of the hinge **317** relative to the base **302**. For example, the handles **343** can be lifted to pivot into an open configuration as shown in FIG. **4A** and rotated back around the pivot axis **P** into the cutting configuration as shown in FIG. **4B**.

[0116] The cartridge **200** may be inserted within the receptacle **306** of the cutting device **300** when the handles **343** are lifted into the open configuration and the cutting member **312** is positioned away from the cutting configuration. As best shown in FIGS. **4D-4E**, the cartridge **200** may be slid into the receptacle **306** to position the recess **221** holding the patch of material **101** below the upper blades **344a**, **344b** and above the lower blades **345a**, **345b**. The cover **214** holding the patch of material **101** within the recess **221** can include an upper portion **220** that tapers into a narrower lower portion **222**. The lower portion **222** of the cover **214** is aligned with the projection **271** of the recess **221** and traps the patch of material **101** therebetween. The upper portion **220** of the cover **214** can slide above the upper blades **344a**, **344b** as the cartridge **200** is installed with the cutting device **300**. The lower portion **222** of the cover **214** is sized to slide between the upper blades **344a**, **344b** as the cartridge **200** is inserted within the receptacle **306** of the cutting device **300**. FIG. **4D** shows the upper blades **344a**, **344b** separated a distance from the lower blades **345a**, **345b** and the

narrow lower portion **222** of the cover **214** positioned between them. FIG. 4E shows the handles **343** rotated back down into the cutting configuration and the upper blades **344a**, **344b** urged downward towards the patch of material **101** and toward the lower blades **345a**, **345b**. The patch of material **101** is cut by the corresponding upper and lower blades forming the stent **105**. The distance between the upper and lower blades determines the width of the stent **105** that is cut from the patch of material **101**.

[0117] The handles **343** can open along any of a number of orientations relative to the base **302**. For example, the pivot axis P of the hinge **317** can be substantially orthogonal to the longitudinal axis of the base A. In this implementation, the hinge **317** can be positioned on a distal end of the base **302** such that the handles **343** hinge open by rotating upward and toward the distal end of the base **302**. The upper blades **344a**, **344b** may be spring-loaded such that they readily return to an open configuration as the handle **343** is lifted or released.

[0118] The stent **105**, once cut, is contained on all sides by the cartridge **200** and the cutting member **312** creating a complete enclosure or stent cutting chamber for the stent **105** within the assembly of the cutting device **300** and the cartridge **200**. For example, the floor and ceiling of the stent cutting chamber can be formed by the lower portion **222** of the cover **214** and the projection **271** of the recess **221**. The walls of the stent cutting chamber can be formed by the upper blades **344a**, **344b**, and the lower blades **345a**, **345b** of the cutting member **312**. Together, the walls of the stent cutting chamber can form a rectangle to help constrain and direct the pusher **320** of the cutting device **300** that is advanced to push the stent **105** from the stent cutting chamber distally into the lumen **238** of the shaft **210**. In an implementation, the stent cutting chamber can be at least partially arced or circular in cross-section. The upper and lower surfaces of the cutting chamber can be curved or non-planar. As an example, the lower portion **222** of the cover **214** can be recessed forming a concavity forming arched ceiling to the cutting chamber. The floor of the cutting chamber formed by the projection **271** may incorporate a corresponding concavity. The arched ceiling and recessed floor of the cutting chamber reduces the amount of open space created around the cut stent **105** relative to the inner walls of the shaft that could otherwise result in the push rod going off-track or allowing the cut stent **105** to divert off the desired path during deployment. Minimizing the air space within the shaft relative to the trephine stent **105** improves advancement of the stent **105** through the device. The cut stent **105**, in turn, can have a cross-sectional shape that conforms more closely to the cross-sectional shape of the delivery conduit through which the stent **105** must be advanced. The corresponding shape eliminates excess space on the upper and lower sides of the cut stent **105** relative to the conduit. This, in turn, provides better guidance for the pusher **320** to advance the cut stent **105** towards the distal end of the shaft. The stent **105** can also be cut to be oversized relative to the conduit as discussed elsewhere herein and compressed, compacted, or otherwise manipulated within the conduit prior to deployment.

[0119] The stent **105**, once cut, can be axially aligned with the lumen **238** of the shaft **210** of the cartridge **200**. FIGS. 4F-4G and also FIGS. 4H-4J show the cutting device **300** can include a pusher **320** configured to slide distally relative to the base **302** into a proximal end region the cartridge **200** to advance the cut stent **105** from the location of this complete enclosure along the implantation conduit into the lumen **238** of the shaft **210**. The pusher **320** is not visible in the implementation of FIG. 6. However, the base **302** can include an actuator **304** such as a dial, button, slider, or other input that is operatively coupled to the pusher **320** that upon actuation causes the pusher **320** to move distally relative to the base **302**. Any of a variety of user actuators **304** are considered herein to move the pusher **320** to prime the stent **105** in place relative to the lumen **238**. This priming step with the pusher **320** of the cutting device **300** ensures the cut stent **105** is held within a fully enclosed space on all sides (i.e. a region of the shaft **210**) after removal of the cartridge **200** from the cutting device **300** and before coupling of the cartridge **200** with the delivery device **400**.

[0120] FIG. 4H shows that while the handles **343** are urged downward towards the base **302** (e.g., the blades **344** positioned in the cutting configuration relative to the implant **105**), the pusher **320**

of the cutting device **300** can be advanced distally through the base **302**. FIG. **4I** shows the pusher **320** ready to engage the stent **105** within the recess **221** on a proximal end. FIG. **4J** shows the pusher **320** has advanced the stent **105** distally into the lumen **238** of the shaft **210** of the cartridge **200**. As mentioned above, the blades **344** besides, the cover **214** above, and the projection **271** below created the complete enclosure for the cut stent **105** on all sides preventing the stent **105** from buckling within the lumen **238** during this distal advancement into the lumen **238**. The conduit within which the stent **105** is held is size-matched (or under-sized) to the outer dimension of the stent being implanted thereby preventing buckling and wrinkling as the stent **105** is urged into the primed position.

[0121] The stent **105** can be urged into a distal end region **212** of the shaft **210** and the cartridge **200** removed from the cutting device **300**. Once the cutting device **300** and the cartridge **200** are disengaged with one another, the cartridge **200** is ready to be loaded with the delivery device **400** to insert the stent **105** into the eye.

[0122] The patch of material **101** can be cut and loaded within the shaft **210** of the cartridge **200** in a variety of ways. As discussed elsewhere, the patch of material **101** can be cut to substantially the same size as the conduit through which it will be delivered. The patch of material **101** can preferably be cut to a size that is slightly larger than the size of the conduit through which it is delivered so that the stent **105** is compressed and packed within the conduit so that it may be more easily advanced through the lumen **238**. The cutting can be performed as described above with respect to FIGS. **4A-4E**. The cutting of the patch of material and transfer into the shaft **210** can also be performed using other cutting assemblies **500** as described below and with respect to FIGS. **14A-14H**. The cutting assemblies **500** described herein can form part of the tissue cartridge **200**, the cutting device **300**, or the delivery device **400**. Preferably, the cutting assembly **500** is part of the cutting device **300**. The cutting device **300** can couple to at least a portion of the cartridge **200** such as the nose cone assembly **274** with the distal shaft **210** extending from the nose cone **275** so that the cut stent **105** can be primed within the shaft **210** for delivery using the delivery device **400**. The cartridge **200** can include a proximal portion **207** configured to hold the patch of material for cutting as shown in FIGS. **2**, **3A-3C**, or **7A-7C** or the removable nose cone **274** and shaft **210** as shown in FIGS. **9A-9D**, **10A** that does not include a proximal portion **207** for holding the patch of material. The cartridge **200** whether configured to hold a patch of material for cutting or not can be a transferrable component that is designed to couple with a cutting assembly, primed with the cut stent, removed from the cutting assembly, and coupled with a delivery device for deployment of the cut stent in the eye.

[0123] FIG. **14A** shows an implementation of a cutting assembly **500**. The cutting assembly can be part of a cutting device **300** configured to engage with a cartridge. The cutting assembly **500** can cut a patch of material **101**, which can be held within the cartridge or within a region of the cutting assembly **500**. The cut stent can be transferred from the cutting assembly **500** into a distal shaft **210** of the cartridge **200** for delivery through the shaft into an eye. The cutting assembly **500** can incorporate a cutting die **511** positioned relative to a slot **507** in a base **509** and a movable member **505** having planar cutting surface **513** coupled to the base **509**. The movable member **505** can be swiveled 90 degrees relative to the base **509** from a first position to a second position. When the movable member **505** is swiveled to its second position the patch of material **101** can be placed against the cutting surface **513**. The cutting die **511** can compress the patch of material **101** against the cutting surface **513**. Advancing the cutting die **511** towards the cutting surface **513** can cut through the patch of material **101** in two locations as described elsewhere herein. The excess tissue can be removed from the cutting surface **513** and the movable member **505** still holding the cut stent **105** on its cutting surface **513** swiveled back towards the first position. This arranges the cut stent **105** on the cutting surface **513** within the path of the slot **507** so that a compacting tool **517** or other member can load the cut stent **105** into the slot **507**. The slot **507** can have a terminal region **508** that aligns with a longitudinal axis A of the distal shaft **210** when the cartridge **200** is coupled

to the cutting device **300**. The terminal region **508** can have a cross-sectional shape that is rounded similar to a cross-sectional shape of the distal shaft **210**. The cut stent **105** positioned within the terminal region **508** can then be urged into the lumen of the distal shaft **210** so that it is primed for delivery. The size of the slot **507** and/or the terminal region **508** can be smaller than the size of the cut stent **105** so that advancement of the compacting tool **517** urging the cut stent **105** into the slot **507** causes the stent **105** to be compressed and compacted into a plug. Once the cut stent **105** is positioned within the distal shaft **210** of the cartridge **200**, the cartridge **200** can be removed from the cutting device **300** and transferred to a delivery device **400** for deployment in the eye.

[0124] FIG. **14B** shows an interrelated implementation of a cutting assembly **500** for cutting the patch of material **101** and transferring the cut stent **105** for delivery. As with the embodiment of FIG. **14A**, the cutting assembly **500** can be part of a cutting device **300** configured to engage with a cartridge. The patch of material can be held within a region of the cartridge for cutting or can be held by a portion of the cutting assembly **500**. The cutting die **511** can insert through a compression pad **515** to cut the patch of material **101**. The patch of material **101** can be positioned against a cutting surface **513**. The cutting surface **513** need not be part of a movable member as in the prior implementation, but can be at least a portion of the base **509**. The patch of material **101** can be compressed between the cutting surface **513** of the base **509** and the compression pad **515**. The cutting die **511** can be advanced through the compression pad **515** so that blades of the cutting die **511** slice through the patch of material **101** in two locations. After the patch of material **101** is cut the excess tissue can be removed and pressure applied by the compression pad **515** released. The cutting die **511** can include a spring **516** so that it returns to its initial position and the pressure pad **515** and cutting die **511** no longer apply a pressure against the cut stent **105**. The cut stent **105** can be positioned relative to a slot **507** in the base **509** so that the compacting tool **517** can urge the cut stent **105** through the slot **507** toward the terminal region **508**. As discussed elsewhere, the cut stent **105** can be oversized relative to the size of the slot **507** so that urging the stent into the conduit compresses and compacts the stent **105** for delivery. The slot **507** can have a terminal region **508** that aligns with a longitudinal axis A of the distal shaft **210** when the cartridge is coupled to the cutting device **300**. The cut stent **105** positioned within the terminal region **508** can then be urged into the distal shaft **210** so that it is primed for delivery. The cartridge, now containing the cut stent **105**, can be removed from the cutting device **300** and transferred to a delivery device **400** for deployment in the eye.

[0125] FIG. **14C** shows an interrelated implementation of a cutting assembly **500** for cutting the patch of material **101** and transferring the cut stent **105** for delivery. The cutting assembly **500** can additionally incorporate a movable stop **520** positioned between the patch of material **101** and the slot **507** through which the cut stent **105** is to be advanced. The compression pad **515** and cutting die **511** can compress the patch of material **101** against a cutting surface **513** of the base **509**. The patch of material **101** can be enclosed between the cutting surface **513** on an underside, the movable stop **520** on a distal side and the compression pad **515** on an upper side. The cutting die **511** can include a single blade and be advanced through the compressed patch of material **101** to cut the patch in a single location creating a stent **105**. The cutting die **511**, compression pad **515**, and movable stop **520** can be withdrawn away from the cut stent **105** so that the compacting tool **517** can urge the cut stent **105** distally into the slot **507** for delivery. The terminal region **508** of the slot **507** can align with a longitudinal axis A of the distal shaft **210** when the cartridge is coupled to the cutting device **300**. The cut stent **105** positioned within the terminal region **508** can then be urged into the distal shaft **210** so that it is primed for delivery as described elsewhere. FIG. **14I** shows a nose cone assembly **274** arranged relative to the cutting assembly **500** of FIG. **14C**. The longitudinal axis A of the distal shaft **210** of the nose cone assembly **274** can be aligned with the terminal region **508** of the slot **507** so that the compacting tool **517** can urge the cut stent **105** into the shaft **210**. Once the cut stent **105** is compacted into the lumen **238** of the shaft **210** the nose cone assembly **274** can be removed from its association with the cutting assembly **500** and

transferred to a delivery device **400** for deployment in the eye.

[0126] The position of the movable stop **520** relative to the cutting blade of the die **511** can be adjusted to achieve different stent widths. For example, the movable stop **520** can be moved toward the single blade of the cutting die **511** to decrease the width of the stent and moved away from the cutting die **511** to increase the width of the stent. The location of the movable stop **520** relative to the cutting die **511** can be selected by a user, for example, via a dial or other user interface that allows for incremental adjustments. The dial range can be between about 0.6 mm and about 1.9 mm and can include markings that are laid out per a 1/4 to 1/16 thread. The cutting die **511** of the cutting assembly **500** can be attached to a lever, handle, or other actuator **343** as described elsewhere herein, to advance the single blade through the patch of material **101** held against the cutting surface **513** by the pad **515** upon selection of the width. In an implementation, the cutting surface **513** can be 1/16" 90A silicone.

[0127] FIGS. **15A-15B** illustrate a cutting device **300** having a cutting assembly **500**. The cutting device **300** can include a handle **543** movably coupled to the base **509** to actuate the cutting assembly **500**. For example, the handle **543** is configured to raise and lower a cutting die **511** relative to the cutting surface **513** of the base **509**. The cutting surface **513** can include a recess **544** sized to hold a patch of material (not shown). The cutting surface **513** can be movable relative to the base **509** to expose the recess **544** for positioning the patch of material **101** within the recess **544**. The cutting device **300** can incorporate an actuator **545** such as a dial, button, slider, switch, or other type of actuator configured to adjust the position of the blade **511** relative to the cutting surface **513** as discussed above. The actuator **545** can move the base **509** side-to-side via a threaded screw or other mechanism to change the position of the patch of material **101** held within the recess **544** relative to the cutting die **511** and thereby modify the width of the stent cut from the patch. Alternatively, the actuator **545** can move the die **511** relative to the recess **544** to change the width of the stent. The cutting device **300** can incorporate a stage **546** configured to be movable relative to the base **509** such as by sliding, swiveling, or lifting away from the base **509**. In some implementations, the stage **546** can slide within a single plane relative to the underlying base **509** while remaining connected to the base **509** at least in part. Alternatively, the stage **546** can be removed entirely from the base **509**. Moving the stage **546** relative to the base **509** can reveal the recess **544** out from under the area of the device where the cutting die **511** and handle **543** are located. This allows for loading of a patch within the recess **544** without the components of the cutting assembly **500** obstructing a user's view or blocking access physically. The cutting device **300** can be a solo cutter and need not incorporate a compression or holding mechanism or a transferring mechanism. Rather, the cut stent **105** following cutting with the cutting assembly **500** can be manually transferred to another tool for priming the cut stent **105** for deployment through a shaft.

[0128] FIG. **14D** shows an interrelated implementation of a cutting assembly **500** for cutting the patch of material **101**. The cutting assembly **500** can include a paper hole punch sort of cutting. Sharp corners or raised sharp edges **525** can project from the cutting surface **513**. The sharp edges **525** can surround a hole **527** through the cutting surface **513** that leads directly into the slot **507** of the base **509**. A patch of material **101** can be positioned against the cutting surface **513** over the hole **527** and against the sharp edges **525**. A punch **511** can be urged against the patch of material **101** from above so that the patch of material **101** is cut by the sharp edges **525** and the cut stent **105** is urged through the hole **527** into the slot **507** by the punch **511**. The cut stent **105** can then be arranged within the slot **507** so that a pusher (not shown in FIG. **14D**) may urge the cut stent **105** through the slot **507** towards the terminal end **508**. The terminal region **508** of the slot **507** aligns the cut stent **105** with the longitudinal axis A of the distal shaft **210** so that the stent can be urged into the distal shaft **210** so that it is primed for delivery. The cartridge can be removed from the cutting device **300** and transferred to a delivery device **400** for deployment in the eye.

[0129] FIG. **14E** shows an interrelated implementation of a cutting assembly **500** for cutting the

patch of material **101**. The cutting assembly **500** can also incorporate a money plunger sort of cutting. The patch of material **101** can be positioned over a slot **507** in a base **509** and a cutting die **511** urged from above against the material **101** so that the cutting edges of the die **511** can slice through the patch of material **101** in two locations to cut the stent **105** to length. A compacting tool **517** can be advanced through a bore **529** in the die **511** to drive the cut stent **105** into the slot **507** urging it to a terminal region **508** of the slot **507**. The compacting tool **517** or an additional compression tool **421** can be advanced through the bore **529** in the die **511** to compress the cut stent **105** within the terminal region **508** of the slot **507** to compact it and align the cut stent **105** with the distal shaft **210** so that it is primed for delivery. The cartridge can be removed from the cutting device **300** and transferred to a delivery device **400** for deployment in the eye.

[0130] FIG. **14F** shows an interrelated implementation of a cutting assembly **500** for cutting the patch of material **101**. The cutting assembly **500** can incorporate forceps-like tool **530** to clamp the patch of material **101**. A scalpel or other cutting tool **535** can be used to trim the patch of material **101** held by the forceps **530** to length. The forceps **530** holding the cut stent **105** can be arranged relative to a base **509** and the clamp pressure of the forceps **530** released. A compacting tool **517** can be advanced through the forceps **530** to urge the cut stent **105** from the forceps **530** into a slot **507** of the base **509** for compressing and compacting the cut stent **105** for delivery as described above.

[0131] FIG. **14G** shows an interrelated implementation of a cutting assembly **500** for cutting the patch of material **101**. The cutting assembly **500** can incorporate a plunger **511** configured to compress a patch of material **101** within a transfer slot **537** of a transfer base **539**. The patch of material **101** can be trimmed to size with a scalpel or other cutting tool **535**. The cut stent **105** within the transfer slot **537** can be transferred by attaching to the transfer base **539** to a base **509** with a defined slot **507** in a manner that aligns transfer slot **537** to slot **507** for compressing and loading of the cut stent **105** using a compacting tool **517** for deployment.

[0132] FIG. **14H** shows an interrelated implementation of a cutting assembly **500** for cutting the patch of material **101**. The cutting assembly **500** can incorporate a rotating cylinder **540** configured to cut and arrange the cut stent **105** relative to a slot **507** in a base **509** for loading and compressing the stent **105** for delivery. The rotating cylinder **540** can incorporate an internal slot **542** for receiving at least a portion of the patch of material **101**. Rotation of the cylinder **540** trims the excess tissue extending beyond the slot **542** in the cylinder **540**. The cut stent **105** trimmed to length within the slot **542** of the cylinder **540** is then arranged relative to the slot **507** in the base **509** for loading and compression for delivery.

[0133] The cut stent **105** that is loaded and compressed for delivery can be positioned within at least a portion of the cartridge **200**, such as within a lumen **238** of the shaft **210**. At least a portion of the cartridge **200** can be removed from the cutting device **300** and engaged with a delivery device **400** for deployment of the stent **105** from the cartridge **200** into the eye. The compression and transfer of the cut stent **105** described above in relation to the cutting assembly **500** prepares the cut stent **105** for delivery without the cut stent **105** being removed from the cartridge **200**.

[0134] The cartridge **200** is described herein as being configured to couple with a cutting device **300** having a cutting assembly **500** for cutting a patch of material **101** and then removed from engagement with the cutting device **300** so that it can be coupled to a delivery device **400**. This relationship can include removing and re-engaging the entire cartridge **200** or just a portion of the cartridge **200**, such as just the nose cone assembly **274** (e.g., the nose cone **275** and the shaft **210**). Both arrangements are considered herein. The nose cone assembly **274** may be referred to herein simply as the cartridge **200**. Where the cartridge **200** is described as removed from the cutting device **300** the description is relevant to just the nose cone assembly **274** being removed or the entire cartridge **200** being removed from the cutting device **300**. Where the cartridge **200** is described as configured to engage with the delivery device **400** that the description is relevant to just the nose cone assembly **274** being engaged or the entire cartridge **200** being engaged to the

delivery device **400**. Each instance of coupling between the cartridge **200** and another component of the system **100** may be the entire cartridge **200** or just a portion of the cartridge **200** such as the nose cone assembly **274**.

[0135] The patch of material **101** can be placed within a portion of the cartridge **200** for cutting or the patch of material **101** can be placed within a portion of the cutting device **300** for cutting by the cutting assembly **500** and the cut stent **105** transferred to the cartridge **200** (or just a portion of the cartridge **200** such as the nose cone assembly **274**). The cut stent **105** can be transferred using a component of the cutting assembly **500** into the cartridge **200**, which is then decoupled from the cutting device for coupling with the delivery device. The patch of material **101** can be placed within a region of the cutting assembly **500** for cutting and then the cut stent **105** manually transferred from the cutting assembly **500** for compacting within a delivery shaft **210**. The cut stent **105** can be transferred using a separate device from the cutting assembly **500** including manually. In an implementation, the system includes a cutting device **300** having a cutting assembly **500**. The cut stent **105** from the cutting assembly **500** can be manually transferred (e.g., by forceps) to a transfer device having a compacting tool **517** to compact the cut stent **105** into a distal shaft **210**. The distal shaft **210** having the cut stent **105** compacted therein can then be coupled to a delivery device **400** for deployment of the cut stent **105** in an eye. The system can have separate cutting, transferring, and delivery devices rather than one or more of the devices being integrated. The cutting assemblies **500** shown in FIGS. **14A-14H** can be part of a cutting device. The transferring components of the cutting assemblies **500** can be integrated with the cutting device or can be a separate transferring device.

[0136] The system **100** can include a delivery device **400** that is configured to couple with at least a portion of the cartridge **200** holding the cut stent **105**. In some implementations, the entire cartridge **200** with the cut stent **105** is removed from the cutting device **300** and engaged with the delivery device **400** (see FIG. **2**). In interrelated implementations, a portion of the cartridge **200** with the cut stent **105** positioned therein is removed from the cutting device **300** and engaged with the delivery device **400** (see FIGS. **6, 9A-9D**).

[0137] In the implementation shown in FIGS. **5A-5B**, the cartridge **200** holding the cut stent **105** can be removed loaded into the delivery device **400**. FIGS. **5C-5F** illustrate loading of the tissue cartridge **200** within the delivery device **400** and deployment of the cut stent **105** using the delivery device **400**. The delivery device **400** together with the cartridge **200** can be used to deliver the stent **105** into the implanted location, such as via an ab interno delivery pathway. This allows for loading the stent and deploying the stent without having to remove the cut stent **105** from its location within the cartridge **200** in order to load the cut stent **105** into the delivery device **400**. At least a portion of the cartridge **200** (e.g., the proximal portion **207** of the cartridge **200** or a region of the nose cone assembly **274**) can be held by the delivery device **400** and the distal portion **205** of the cartridge **200** can be inserted into the eye.

[0138] The delivery device **400** can include a proximal handle **405** that is sized and shaped to be grasped by a single hand of a user and a distal end region **410** defining an attachment mechanism **425** such as a receptacle **412** sized to engage with at least a portion of the cartridge **200**. In an implementation, the receptacle **412** can be sized to receive at least a length of the proximal portion **207** of the cartridge **200** (see FIG. **5C**). In an interrelated implementation, the attachment mechanism **425** can incorporate another male-to-female attachment mechanism such as a bayonet connection **413** (see FIGS. **10A-10C**). As mentioned above with respect to the cutting device **300**, the attachment mechanism **425** can be keyed such that the cartridge **200** with the cover **214** in place on the base **224** can be received within or otherwise engage the attachment mechanism **425** in a single orientation. When the cartridge **200** is coupled with the attachment mechanism **425** of the handle **405**, the shaft **210** of the cartridge **200** extends in a distal direction outward from the handle **405**. The keying features of the attachment mechanism **425** can prevent attachment in the wrong orientation. The attachment mechanism **425** can also provide a secure connection with tactile

feedback to the user to indicate when the connection is fully engaged. The attachment mechanism **425** also is dimensioned to ensure alignment of the lumen **238** of the shaft **210** with the internal mechanisms of the delivery device **400** such as the push rod **420**.

[0139] The attachment mechanism **425** of FIGS. 5A-5C can be a receptacle **412** having a depth sufficient to contain a length of the proximal portion **207** of the cartridge **200** while the shaft **210** remains outside the receptacle **412**. A flexible hook **422** can extend into at least a portion of the receptacle **412** (see FIG. 5C). A distal end **424** of the hook **422** can be received within a correspondingly shaped detent **272** near a proximal end region of the tissue cartridge **200**. As the cartridge **200** slides within the receptacle **412**, the distal end **424** of the hook **422** can slide through the proximal end **207** of the cartridge **200** and insert within the detent **272**. The flexibility of the hook **422** allows for the hook **422** to be urged upward as the distal end **424** of the hook **422** is advanced through a first region of the cartridge **200** and flex back downward as the distal end **424** is advanced further to thereby engage the detent **272** (see FIG. 5D). The spring-loaded hook **422** engaging with the detent **272** can provide a tactile and/or auditory “click” to inform a user that the cartridge **200** is fully installed within the delivery device **400**, retained and ready for delivery of the stent **105**.

[0140] One or more actuators **415** can be positioned on a region of the handle **405**. The actuator **415** can also be manipulated by the single hand of the user such as with a thumb or finger. The configuration of the actuator **415** can vary. For example, the actuator **415** can include any of a variety of knob, button, slider, dial, or other type of actuator configured to move one or more components of the delivery device **400** as will be described in more detail below.

[0141] The delivery device **400** can include a compacting tool **420** configured to be moved by the one or more actuators **415**. The compacting tool **420** can be used together with the cartridge **200** to deliver the stent **105** from the cartridge **200** once the desired position is reached with the distal end of the shaft **210**. The compacting tool **420** can be sized and shaped complementary to the inner dimension of the shaft **210**. For example, where the shaft **210** of the cartridge **200** has a rectangular cross-sectional shape, the compacting tool **420** may be rectangular in cross-section. This allows the compacting tool **420** to effectively urge the cut stent **105** through the lumen **238** of the shaft **210**.

[0142] The compacting tool **420** can be fully retracted in a proximal position prior to coupling of the tissue cartridge **200** within the delivery device **400** so the compacting tool **420** does not interfere with loading of the cartridge **200**. Once the cartridge **200** is installed and retained within the delivery device **400** as shown in FIG. 5D and FIG. 9B, the compacting tool **420** can be advanced distally through a proximal port in the cartridge **200** and into the lumen **238** of the shaft **210** (see FIG. 5E and FIG. 9C). In some implementations, the compacting tool **420** can be advanced through the lumen **238** and out the distal opening **230** from the lumen **238** to deploy the stent **105**. In other implementations, the compacting tool **420** is advanced to a distal location near the proximal end of the stent **105** within the lumen **238** and the shaft **210** is withdrawn proximally while the compacting tool **420** remains stationary to deploy the stent **105** (see FIG. 5F and FIG. 9D).

[0143] The shaft **210** can be withdrawn proximally via motion of the cartridge **200** in a proximal direction relative to the delivery device **400** while the compacting tool **420** remains stationary in order to deploy the stent **105** within the eye (see FIG. 5F and FIG. 9D). The compacting tool **420** therefore can act as a stopper thereby preventing the stent **105** from following the shaft **210** as it is retracted. The result is that the stent **105** is unsheathed from the shaft **210** and left within the tissues. In other implementations, both the cartridge **200** and the compacting tool **420** are movable to effect deployment of the stent from the shaft **210**.

[0144] In some implementations, the compacting tool **420** can be coupled to a first actuator **415** and the cartridge **200** can be coupled to a second actuator **415**. The first and second actuators **415** can be sliders, buttons, or other configuration or combination of actuators configured to advance and retract their respective components. The first actuator **415** coupled to the compacting tool **420**

can be withdrawn proximally such that the compacting tool **420** is in its most proximal position when the cartridge **200** is engaged by the attachment mechanism **425** of the delivery device **400**. The user can advance the first actuator **415** to urge the compacting tool **420** distally to advance the stent **105** within the lumen **238** of the cartridge **200** towards the distal opening **230** of the shaft **210**. After the cut stent **105** is primed into its distal position within the lumen **238**, the shaft **210** of the cartridge **200** can be used to dissect tissue of the eye until a target location is accessed. Once the shaft **210** is in position to deploy the stent **105** in the eye, the first actuator **415** coupled to the compacting tool **420** can be maintained in this distal position and the second actuator **415** actuated (e.g., withdrawing a slider or pushing a button) to retract the cartridge **200** a distance relative to the delivery device **400**. This relative movement of the shaft **210** of the cartridge **200** to the compacting tool **420** deploys the stent **105** from the lumen **238** in the anatomy.

[0145] FIG. 5E shows the cartridge **200** installed within the receptacle **412** of the delivery device **400** such that a space exists between the terminal end of the receptacle **412** and the proximal-most end of the cartridge **200**. The depth of this space defines the maximum distance the cartridge **200** can be retracted. The stent **105** is located near the distal opening **230** from the lumen **238** and the compacting tool **420** is advanced to its distal position such that the distal end of the compacting tool **420** abuts against a proximal end of the stent **105**. The distal end **424** of the hook **422** is retained within the detent **272** and the second actuator **415** is not yet actuated. A proximal end **426** of the hook **422** is coupled to a spring **430**. When the second actuator **415** is in a resting state prior to actuation, the hook **422** is urged distally into a first configuration. The spring **430** is compressed between the proximal end **426** of the hook **422** and the distal end of the spring **430** housing when the hook **422** is urged distally into the first configuration. When the second actuator **415** is actuated (e.g., pushed downward), the spring **430** is released and urges the proximal end **426** of the hook **422** towards a proximal end of the handle **405**. The hook **422** moves proximally and drags along with it the cartridge **200**, which is coupled to the hook **422** due to engagement of the distal end **424** of the hook **422** within the detent **272**. The distance the hook **422** moves proximally thus, retracts the cartridge **200** deeper into the receptacle **412**. The compacting tool **420** can remain stationary during cartridge **200** retraction. The relative motion between the shaft **210** and the compacting tool **420** deploys the stent **105** from the lumen **238** (see FIG. 5F).

[0146] It should be appreciated that additional distal movement of the compacting tool **420** can be used to aid in deployment of the stent **105** from the lumen **238**. It should also be appreciated that compacting tool **420** advancement and cartridge **200** retraction can be controlled by dual actuators **415** as described above or by a single actuator **415** capable of both pusher and cartridge **200** movement depending on degree of actuation. Additionally, the shaft **210** can be used to inject a viscous material such as viscoelastic during the procedure using the compacting tool **420** as a plunger. The methods of implantation and delivery of the stent **105** are described in more detail below.

[0147] FIGS. 11A-11C illustrate steps in the deployment of the stent using a first actuator **415a**, which in this case can be a slider, of the delivery device **400** to move the pusher from a first loading position (fully retracted) to a second primed position (at least partially advanced). The first loading position retracts the pusher away from the distal end region of the delivery device **400** allowing the nose cone assembly **274** (or entire cartridge **200**) to be coupled to the delivery device **400**. The second, primed position advances the pusher towards the distal end of the delivery device **400** to advance the cut stent **105** through the lumen **238** of the shaft **210**. Preferably, the pusher is advanced to the second, primed position prior to insertion of the shaft **210** through the cornea. The delivery device **400** can additionally incorporate a movable guard **432** arranged to prevent a user from inadvertently pushing the slider beyond the second primed position. The guard **432** can be pushed down toward the housing of the delivery device so that a second actuator **415b** is covered by the guard **432** preventing the second actuator **415b** from being inadvertently activated. The guard **432** has a length so that the guard **432** extends over at least a portion of the slider track

thereby blocking the first actuator **415a** from moving further distal in addition to blocking the second actuator **415b** (FIG. 11B). Once the stent **105** is advanced to the primed position and is ready to be deployed in the eye, the guard **432** can be rotated up out of the way revealing the second actuator **415b**. The first actuator **415a** is free to slide further distal along the track and the second actuator **415b** is available to be depressed (FIG. 11C). The guard **432** can also be fully removed from the device **400** or the device **400** not include any guard **432**. The housing of the device **400** can include one or more marks **434** intended to provide feedback to a user regarding the position of the compacting tool **420** through the shaft **210**. The advancement of the compacting tool **420** into one or more positions relative to the housing can also provide tactile feedback to a user as described elsewhere herein.

[0148] FIGS. 12A-12D illustrate the delivery device **400** in cross-section prior to advancing the pusher rod **420** to the second position and after advancing the pusher rod **420** to the second position. Once the nose cone assembly **274** is attached to the delivery device **400**, the first actuator **415** and the compacting tool **420** can be advanced from the initial, retracted first position to a second position. The first actuator **415a** and the compacting tool **420** can be advanced to the second position causing the compacting tool **420** to insert into the lumen **238** behind the material to be delivered (e.g. cut stent **105**). The guard **432** can prevent the first actuator **415a** from sliding beyond the second position. The location of the second position is designed to place the leading face of the compacting tool **420** a predefined distance away from the distal tip of the shaft **210** (e.g. 6 mm). Once the user has created the desired cleft and are ready to deliver material from the lumen, the compacting tool **420** can be advanced to its third, forward-most position (with the guard **432** out of the way or otherwise removed or absent from the device **400**). The second actuator **415b** can be engaged to release the material from the shaft **210**. The second actuator **415b** as described elsewhere herein can retract the shaft **210** while the compacting tool **420** remains fixed ultimately releasing the stent **105** from the lumen. The nose cone assembly **274** withdraws and the compacting tool **420** stays fixed.

[0149] The delivery device **400** and the cartridge **200** (or nose cone assembly **274**) can be single use devices or may be sterilized and re-used. FIGS. 13A-13B illustrate a reset mechanism **436** so that the deployment structures can be reset and the delivery device **400** may be re-used. Activating the reset mechanism **436**, for example sliding a button forward, can return the deployment structures to an armed position. The reset mechanism **436** may also be performed by pulling on the nose cone assembly **274** or the bayonette connector **413** of the delivery device **400** distally until the second actuator **415** returns to its original armed position. The nose cone assembly **274** can be removed from the delivery device **400**, if desired, and additional material loaded into the shaft **210** as described elsewhere herein. The delivery device **400** may be provided in an actuated or unarmed state and a user arm the instrument at the time of use.

[0150] A nose cone assembly transferable between the delivery device and the cutting device can be mounted relative to a main assembly of the cutting device. A patch of tissue can be cut by the cutting device and loaded into the nose cone assembly, which in turn, can be transferred from the main assembly of the cutting device back to being coupled with the delivery device for use in deployment in a patient. The configuration of the nose cone assembly can vary including any of the transferrable cartridges described herein. In an implementation, the nose cone assembly may be mounted relative to a cutting assembly by coupling a proximal end of the nose cone to the base such that a longitudinal axis of the lumen of the shaft extending distally from the nose cone aligns with a longitudinal axis of a corresponding conduit out from the slot. A tissue patch can be placed within a loading zone area of the base relative to a movable stopper plate on the main assembly. The loading zone area and movable stopper plate may both be part of the base of the main assembly. The patch can be laid inside of one or more alignment features of the loading zone and slid forward into a cutting zone until the patch abuts the stopper plate. Once positioned against the stopper plate, the tissue patch is positioned a specified width by the cutter. Thus, the stopper plate

provides a calibrated stopping point for the tissue patch prior to cutting. An element designed to fix the tissue patch in this position can be activated such as being lowered down over the tissue patch to hold the tissue in place and optionally compress the tissue to a specific height prior to cutting. Once this holding plate is lowered down onto the patch to hold it in place, the cutting lever can be lowered to cut the tissue patch with one or more blades. The stopper plate and holding plate can be moved away from the cut stent and the remainder of the tissue patch removed from the assembly. The cut stent can be loaded using a tissue loader slider. The tissue loader slider can urge the cut stent into position relative to the longitudinal axis of the shaft in the nose cone assembly. For example, the tissue loader slider can be put into place and slid as far forward as possible until the slider abuts a ledge on the main assembly indicating that the cut stent has been fully delivered into the compression channel and is ready to be advanced into the shaft of the nose cone assembly. An elongate tool such as a tissue advancer rod can be inserted into the main assembly along the longitudinal axis to urge the cut stent from the main assembly into the shaft of the nose cone assembly. The rod can be designed to advance the tissue slide towards the tip of the nose cone assembly without pushing the cut stent entirely out of the lumen of the shaft. The nose cone assembly can then be disconnected from the main assembly and attached to a delivery device for deployment into a patient.

[0151] In other implementations the cartridge **200** itself holds the patch of tissue for cutting. For example, FIG. 3A shows the cover **214** of the cartridge **200** can be removed from the slot **214** in the base **224** revealing the recess **221**. A patch of material **101** may be manually loaded within the recess **221**. The patch of material **101** may be sized to be received within the recess **221** or may be trimmed to ensure it is sized to be received within the recess **221**. The cover **214** of the cartridge **200** is replaced onto the base **224** and advanced through the slot **215** until the lower portion **222** of the cover **214** engages the patch of material **101** trapping it against the projection **271**. The cover **214** can compress and/or tension the patch of material **101** within the cartridge **200** when in the closed configuration. FIG. 2 shows the loaded tissue cartridge **200** can be installed into the receptacle **306** of the cutting device **300** with the handles **343** in the open configuration. Once installed, the cutting member **312** can be actuated by lowering the handles **343** towards the base **302** thereby urging the blades **344** towards the patch of material **101** until the blades **344** of the cutting member **312** fully slice through the patch of material **101** (FIG. 4B). With the blades **344** still in the full cut position relative to the cartridge **200**, the pusher **320** of the cutting device **300** can be urged distally to prime the shaft **210** and place the now cut stent **105** within the lumen **238** of the shaft **210** towards the opening **230** from the lumen **238** near the distal end region **212** of the shaft **210**. The pusher **320** can be retracted from the cartridge **200** and the cartridge **200** removed from the cutting device **300**. As described elsewhere herein, removal of the cartridge **200** from the cutting device **300** can include removing the entire cartridge **200** from the device **300** or detaching a nose cone assembly **274** of the cartridge **200** as shown in FIG. 6.

[0152] The primed tissue cartridge **200** having the cut stent **105** positioned within the lumen **238** of the shaft **210** can be installed with the delivery device **400** (e.g., inserted within the receptacle **412** or attached by a bayonet connector **413** or other attachment mechanism **425**). The compacting tool **420** of the delivery device **400** is withdrawn in the proximal-most position and the cartridge **200** coupled to the delivery device **400**. The compacting tool **420** can be advanced using a first actuator **415** from the first, retracted position suitable for loading the cartridge **200** to a second primed position so that the delivery device **400** and cartridge **200** are now ready to be used on a patient.

[0153] In general, the stent **105** positioned within the shaft **210** can be implanted through a clear corneal or scleral incision that is formed using the shaft **210** or a device separate from the cartridge **200**. A viewing lens such as a gonioscopy lens can be positioned adjacent the cornea. The viewing lens enables viewing of internal regions of the eye, such as the scleral spur and scleral junction, from a location in front of the eye. The viewing lens may optionally include one or more guide channels sized to receive the shaft **210**. An endoscope can also be used during delivery to aid in

visualization. Ultrasonic guidance can be used as well using high-resolution bio-microscopy, OCT, and the like. Alternatively, a small endoscope can be inserted through another limbal incision in the eye to image the eye during implantation.

[0154] The distal tip **216** of the shaft **210** can penetrate through the cornea (or sclera) to access the anterior chamber. In this regard, the single incision can be made in the eye, such as within the limbus of the cornea. In an embodiment, the incision is very close to the limbus, such as either at the level of the limbus or within 2 mm of the limbus in the clear cornea. The shaft **210** can be used to make the incision or a separate cutting device can be used. For example, a knife-tipped device or diamond knife can be used initially to enter the cornea. A second device with a spatula tip can then be advanced over the knife tip wherein the plane of the spatula is positioned to coincide with the dissection plane. The spatula tip device can be the shaft **210**.

[0155] The corneal incision can have a size that is sufficient to permit passage of the shaft **210**. In an embodiment, the incision is about 1 mm in size. In another embodiment, the incision is no greater than about 2.85 mm in size. In another embodiment, the incision is no greater than about 2.85 mm and is greater than about 1.5 mm. It has been observed that an incision of up to 2.85 mm is a self-sealing incision.

[0156] After insertion through the incision, the shaft **210** can be advanced into the anterior chamber along a pathway that enables the stent **105** to be delivered from the anterior chamber into the target location, such as the supraciliary or suprachoroidal space. With the shaft positioned for approach, the shaft **210** can be advanced further into the eye such that the distal-most tip **216** of the shaft **210** penetrates the tissue at the angle of the eye, for example, the iris root or a region of the ciliary body or the iris root part of the ciliary body near its tissue border with the scleral spur.

[0157] The scleral spur is an anatomic landmark on the wall of the angle of the eye. The scleral spur is above the level of the iris but below the level of the trabecular meshwork. In some eyes, the scleral spur can be masked by the lower band of the pigmented trabecular meshwork and be directly behind it. The shaft **210** can travel along a pathway that is toward the angle of the eye and the scleral spur such that the shaft **210** passes near the scleral spur on the way to the supraciliary space, but does not necessarily penetrate the scleral spur during delivery. Rather, the shaft **210** can abut the scleral spur and move downward to dissect the tissue boundary between the sclera and the ciliary body, the dissection entry point starting just below the scleral spur near the iris root or the iris root portion of the ciliary body. In another embodiment, the delivery pathway of the implant intersects the scleral spur.

[0158] The shaft **210** can approach the angle of the eye from the same side of the anterior chamber as the deployment location such that the shaft **210** does not have to be advanced across the iris. Alternately, the shaft **210** can approach the angle of the eye from across the anterior chamber AC such that the shaft **210** is advanced across the iris and/or the anterior chamber toward the opposite angle of the eye. The shaft **210** can approach the angle of the eye along a variety of pathways. The shaft **210** does not necessarily cross over the eye and does not intersect the center axis of the eye. In other words, the corneal incision and the location where the stent **105** is implanted at the angle of the eye can be in the same quadrant when viewed looking toward the eye along the optical axis. Also, the pathway of the stent **105** from the corneal incision to the angle of the eye ought not to pass through the centerline of the eye to avoid interfering with the pupil.

[0159] The shaft **210** can be continuously advanced into the eye, for example approximately 6 mm. The dissection plane of the shaft **210** can follow the curve of the inner scleral wall such that the stent **105** mounted in the shaft, for example after penetrating the iris root or the iris root portion of the ciliary body CB, can bluntly dissect the boundary between tissue layers of the scleral spur and the ciliary body CB such that a distal region of the stent **105** extends through the supraciliary space and then, further on, is positioned between the tissue boundaries of the sclera and the choroid forming the suprachoroidal space.

[0160] Once properly positioned, the stent **105** can be released from the shaft **210**. In some

implementations, the stent **105** can be released by withdrawing the shaft **210** while the compacting tool **420** prevents the stent **105** from withdrawing with the shaft **210**.

[0161] Once implanted, the stent **105** forms a fluid communication pathway between the anterior chamber and the target pathway (e.g., supraciliary space or suprachoroidal space). As mentioned, the stent **105** is not limited to being implanted into the suprachoroidal or supraciliary space. The stent **105** can be implanted in other locations that provide fluid communication between the anterior chamber and locations in the eye, such as Schlemm's Canal or a subconjunctival location of the eye. In another implementation, the stent **105** is implanted to form a fluid communication pathway between the anterior chamber and the Schlemm's Canal and/or communication pathway between the anterior chamber and a subconjunctival location of the eye. It should be appreciated the device described herein can also be used to deliver a stent trans-sclerally as well from an ab interno approach.

[0162] As mentioned above, the material used to form the stent can be impregnated with one or more therapeutic agents for additional treatment of an eye disease process.

[0163] A wide variety of systemic and ocular conditions such as inflammation, infection, cancerous growth, may be prevented or treated using the stents described herein. More specifically, ocular conditions such as glaucoma, proliferative vitreoretinopathy, diabetic retinopathy, uveitis, keratitis, cytomegalovirus retinitis, cystoid macular edema, herpes simplex viral and adenoviral infections can be treated or prevented.

[0164] The following classes of drugs could be delivered using the devices of the present invention: antiproliferatives, antifibrotics, anesthetics, analgesics, cell transport/mobility impeding agents such as colchicine, vincristine, cytochalasin B and related compounds; antiglaucoma drugs including beta-blockers such as timolol, betaxolol, atenolol, and prostaglandin analogues such as bimatoprost, travoprost, latanoprost etc; carbonic anhydrase inhibitors such as acetazolamide, methazolamide, dichlorphenamide, diamox; and neuroprotectants such as nimodipine and related compounds. Additional examples include antibiotics such as tetracycline, chlortetracycline, bacitracin, neomycin, polymyxin, gramicidin, oxytetracycline, chloramphenicol, gentamycin, and erythromycin; antibacterials such as sulfonamides, sulfacetamide, sulfamethizole and sulfisoxazole; anti-fungal agents such as fluconazole, nitrofurazone, amphotericin B, ketoconazole, and related compounds; anti-viral agents such as trifluorothymidine, acyclovir, ganciclovir, DDI, AZT, foscarnet, vidarabine, trifluorouridine, idoxuridine, ribavirin, protease inhibitors and anti-cytomegalovirus agents; antiallergenics such as methapyriline; chlorpheniramine, pyrilamine and propylpyridamine; anti-inflammatories such as hydrocortisone, dexamethasone, fluocinolone, prednisone, prednisolone, methylprednisolone, fluorometholone, betamethasone and triamcinolone; decongestants such as phenylephrine, naphazoline, and tetrahydrazoline; miotics and anticholinesterases such as pilocarpine, carbachol, di-isopropyl fluorophosphate, phospholine iodine, and demecarium bromide; mydriatics such as atropine sulfate, cyclopentolate, homatropine, scopolamine, tropicamide, cyclopentolate; sympathomimetics such as epinephrine and vasoconstrictors and vasodilators; Ranibizumab, Bevacizumab, and Triamcinolone.

[0165] Non-steroidal anti-inflammatories (NSAIDs) may also be delivered, such as cyclooxygenase-1 (COX-1) inhibitors (e.g., acetylsalicylic acid, for example ASPIRIN® from Bayer AG, Leverkusen, Germany; ibuprofen, for example ADVIL® from Wyeth, Collegeville, Pa.; indomethacin; mefenamic acid), COX-2 inhibitors (CELEBREX® from Pharmacia Corp., Peapack, N.J.; COX-1 inhibitors), including a prodrug Nepafenac®; immunosuppressive agents, for example Sirolimus (RAPAMUNE®, from Wyeth, Collegeville, Pa.), or matrix metalloproteinase (MMP) inhibitors (e.g., tetracycline and tetracycline derivatives) that act early within the pathways of an inflammatory response. Anticlotting agents such as heparin, antifibrinogen, fibrinolysin, anticlotting activase, etc., can also be delivered.

[0166] Antidiabetic agents that may be delivered using the present devices include acetohexamide, chlorpropamide, glipizide, glyburide, tolazamide, tolbutamide, insulin, aldose reductase inhibitors,

etc. Some examples of anti-cancer agents include 5-fluorouracil, adriamycin, asparaginase, azacitidine, azathioprine, bleomycin, busulfan, carboplatin, carmustine, chlorambucil, cisplatin, cyclophosphamide, cyclosporine, cytarabine, dacarbazine, dactinomycin, daunorubicin, doxorubicin, estramustine, etoposide, etretinate, filgrastin, floxuridine, fludarabine, fluorouracil, fluoxymesterone, flutamide, goserelin, hydroxyurea, ifosfamide, leuprolide, levamisole, lomustine, nitrogen mustard, melphalan, mercaptopurine, methotrexate, mitomycin, mitotane, pentostatin, pipobroman, plicamycin, procarbazine, sargramostin, streptozocin, tamoxifen, taxol, teniposide, thioguanine, uracil mustard, vinblastine, vincristine and vindesine.

[0167] Hormones, peptides, nucleic acids, saccharides, lipids, glycolipids, glycoproteins, and other macromolecules can be delivered using the present devices. Examples include: endocrine hormones such as pituitary, insulin, insulin-related growth factor, thyroid, growth hormones; heat shock proteins; immunological response modifiers such as muramyl dipeptide, cyclosporins, interferons (including α , β , and γ interferons), interleukin-2, cytokines, FK506 (an epoxy-pyrido-oxaazcyclotricosine-tetrone, also known as Tacrolimus), tumor necrosis factor, pentostatin, thymopentin, transforming factor beta2, erythropoietin; antineogenesis proteins (e.g., anit VEGF, Interfurons), among others and anticlotting agents including anticlotting activase. Further examples of macromolecules that can be delivered include monoclonal antibodies, brain nerve growth factor (BNGF), celiary nerve growth factor (CNGF), vascular endothelial growth factor (VEGF), and monoclonal antibodies directed against such growth factors. Additional examples of immunomodulators include tumor necrosis factor inhibitors such as thalidomide.

[0168] In various implementations, description is made with reference to the figures. However, certain implementations may be practiced without one or more of these specific details, or in combination with other known methods and configurations. In the description, numerous specific details are set forth, such as specific configurations, dimensions, and processes, in order to provide a thorough understanding of the implementations. In other instances, well-known processes and manufacturing techniques have not been described in particular detail in order to not unnecessarily obscure the description. Reference throughout this specification to “one embodiment,” “an embodiment,” “one implementation,” “an implementation,” or the like, means that a particular feature, structure, configuration, or characteristic described is included in at least one embodiment or implementation. Thus, the appearance of the phrase “one embodiment,” “an embodiment,” “one implementation,” “an implementation,” or the like, in various places throughout this specification are not necessarily referring to the same embodiment or implementation. Furthermore, the particular features, structures, configurations, or characteristics may be combined in any suitable manner in one or more implementations.

[0169] The use of relative terms throughout the description may denote a relative position or direction. For example, “distal” may indicate a first direction away from a reference point. Similarly, “proximal” may indicate a location in a second direction opposite to the first direction. The reference point used herein may be the operator such that the terms “proximal” and “distal” are in reference to an operator using the device. A region of the device that is closer to an operator may be described herein as “proximal” and a region of the device that is further away from an operator may be described herein as “distal”. Similarly, the terms “proximal” and “distal” may also be used herein to refer to anatomical locations of a patient from the perspective of an operator or from the perspective of an entry point or along a path of insertion from the entry point of the system. As such, a location that is proximal may mean a location in the patient that is closer to an entry point of the device along a path of insertion towards a target and a location that is distal may mean a location in a patient that is further away from an entry point of the device along a path of insertion towards the target location. However, such terms are provided to establish relative frames of reference, and are not intended to limit the use or orientation of the devices to a specific configuration described in the various implementations.

[0170] As used herein, the term “about” means a range of values including the specified value,

which a person of ordinary skill in the art would consider reasonably similar to the specified value. In aspects, about means within a standard deviation using measurements generally acceptable in the art. In aspects, about means a range extending to $\pm 10\%$ of the specified value. In aspects, about includes the specified value.

[0171] While this specification contains many specifics, these should not be construed as limitations on the scope of what is claimed or of what may be claimed, but rather as descriptions of features specific to particular embodiments. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or a variation of a sub-combination. Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Only a few examples and implementations are disclosed. Variations, modifications and enhancements to the described examples and implementations and other implementations may be made based on what is disclosed.

[0172] In the descriptions above and in the claims, phrases such as “at least one of” or “one or more of” may occur followed by a conjunctive list of elements or features. The term “and/or” may also occur in a list of two or more elements or features. Unless otherwise implicitly or explicitly contradicted by the context in which it is used, such a phrase is intended to mean any of the listed elements or features individually or any of the recited elements or features in combination with any of the other recited elements or features. For example, the phrases “at least one of A and B;” “one or more of A and B;” and “A and/or B” are each intended to mean “A alone, B alone, or A and B together.” A similar interpretation is also intended for lists including three or more items. For example, the phrases “at least one of A, B, and C;” “one or more of A, B, and C;” and “A, B, and/or C” are each intended to mean “A alone, B alone, C alone, A and B together, A and C together, B and C together, or A and B and C together.”

[0173] Use of the term “based on,” above and in the claims is intended to mean, “based at least in part on,” such that an unrecited feature or element is also permissible.

[0174] The systems disclosed herein may be packaged together in a single package. The finished package would be sterilized using sterilization methods such as Ethylene oxide or radiation and labeled and boxed. Instructions for use may also be provided in-box or through an internet link printed on the label.

Claims

1. A method of treating an eye of a patient with an implant of minimally-modified biological tissue, the method comprising: positioning an implant within a lumen of a shaft, the shaft extending from a distal end of a tissue cartridge; coupling at least a portion of the tissue cartridge to a delivery device; inserting a distal end region of the shaft into an anterior chamber of the eye; positioning the distal end region adjacent a target eye tissue; and actuating the delivery device to deploy the implant from the tissue cartridge through at least a portion of the lumen such that the implant engages the target eye tissue.
2. The method of claim 1, wherein the implant is cut from a patch of scleral tissue to have a first size and wherein the implant has a second, smaller size while positioned within the lumen of the shaft.
3. A method of treating an eye with minimally-modified biological tissue.

- 4.** The method of claim 3, wherein the biological tissue is scleral tissue, wherein minimally-modifying the scleral tissue comprises reducing the scleral tissue from a first size into a second, smaller size within a distal shaft.
 - 5.** The method of claim 4, wherein the distal shaft is sized and shaped to be inserted through a self-sealing incision in a cornea of the eye into the anterior chamber.
 - 6.** The method of claim 5, further comprising deploying the scleral tissue in the second, smaller size from the distal shaft between tissue layers near the iridocorneal angle.
 - 7.** The method of claim 6, wherein the scleral tissue in the second, smaller size deployed from the distal shaft returns towards the first size.
 - 8.** The method of claim 7, further comprising treating glaucoma with the deployed scleral tissue.
 - 9.** The method of claim 6, wherein deploying the scleral tissue from the distal shaft between tissue layers near the iridocorneal angle comprises deploying the scleral tissue at least in part within Schlemm's canal and at least in part within the anterior chamber, or at least in part between a ciliary body and sclera of an eye, or at least in part within a cyclodialysis cleft.
 - 10.** The method of claim 6, wherein deploying the scleral tissue from the distal shaft between tissue layers near the iridocorneal angle comprises retracting the distal shaft while maintaining the scleral tissue position relative to the tissue layers.
 - 11.** The method of claim 6, wherein deploying the scleral tissue from the distal shaft between tissue layers near the iridocorneal angle comprises pushing the scleral tissue out of the distal shaft and into position between the tissue layers.
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